Transactions on Information Systems

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Journal:	*Transactions on Information Systems*
Manuscript ID	TOIS-2026-0132
Manuscript Type:	Paper
Date Submitted by the Author:	22-Feb-2026
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Computing Classification Systems:	Information systems~Retrieval models and ranking, Information systems~Retrieval effectiveness, Theory of computation~Automated reasoning

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The Canonical Decomposition of Reciprocal Rank Fusion: Formal Derivation, Properties, and Machine-Verified Proofs for Multi-Modal Multi-Scope Retrieval

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We present the canonical decomposition of Reciprocal Rank Fusion (RRF) into a four-component weighted scoring function for retrieval systems that operate over two modalities (semantic and lexical) and two scopes (primary document and cross-reference). Starting from five axioms characterising rank-based fusion and the harmonic constraint that uniquely determines the RRF kernel $f(r) = 1/(k+r)$, we derive the four-component formula through three independent, commutative extensions: per-ranker weighting, multi-modal instantiation, and multi-scope expansion with min-rank cross-reference aggregation. We prove a uniqueness theorem establishing that any rank fusion function satisfying these axioms and structural constraints must be a reparameterisation of our formula, with the smoothing constant $k > 0$ and four positive component weights $\boldsymbol{\alpha} \in \mathbb{R}^4_{>0}$ as the only free parameters. We establish eight formal properties of the scoring function—non-negativity, upper boundedness, strict monotonicity, convexity (diminishing returns), scale invariance, min-rank semilattice structure, consensus reward, and Lipschitz continuity in the weight space—providing complete proofs for each. Five of these properties have been independently machine-verified in the Lean 4 theorem prover using only core tactics, with no Mathlib dependency. We connect the four-component formula to Condorcet voting theory, showing it constitutes a weighted multi-criteria positional scoring rule, and demonstrate its behaviour through worked numerical examples with sensitivity analysis over weight vectors and the smoothing constant.

CCS Concepts: • **Information systems** → **Retrieval models and ranking**; *Retrieval effectiveness*; • **Theory of computation** → *Automated reasoning*.

Additional Key Words and Phrases: reciprocal rank fusion, rank aggregation, hybrid retrieval, formal verification, multi-modal search

ACM Reference Format:
Mergim Gashi. 2026. The Canonical Decomposition of Reciprocal Rank Fusion: Formal Derivation, Properties, and Machine-Verified Proofs for Multi-Modal Multi-Scope Retrieval. 1, 1 (February 2026), 37 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

Reciprocal Rank Fusion (RRF), introduced by Cormack, Clarke, and Buettcher [4], has become the dominant unsupervised method for combining the outputs of multiple rankers in information retrieval. Given a set of rankers $\mathcal{R} = \{r_1, \ldots, r_m\}$, RRF assigns each document d the score

$$\mathrm{RRF}(d) = \sum_{r \in \mathcal{R}} \frac{1}{k + \mathrm{rank}_r(d)}, \tag{1}$$

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XXXX-XXXX/2026/2-ART $15.00
https://doi.org/10.1145/nnnnnnn.nnnnnnn

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where $\text{rank}_r(d)$ is the rank of d under ranker r and $k > 0$ is a smoothing constant (conventionally $k = 60$). Despite its simplicity, RRF consistently outperforms both individual rankers and more complex aggregation methods such as Condorcet voting and CombMNZ [1, 4], and has been widely adopted in production retrieval systems including Elasticsearch [7].

The recent convergence of three trends in retrieval system design motivates a principled extension of the RRF framework. First, *hybrid retrieval* systems routinely combine dense semantic search (via learned embeddings [10]) with sparse lexical search (via BM25 [14]), creating a natural two-modality structure. Second, *retrieval-augmented generation* (RAG) pipelines [11] increasingly exploit document metadata, citation graphs, and cross-reference structures as additional evidence for relevance, introducing a second dimension of scope beyond the document itself. Third, practitioners have introduced per-ranker weighting to allow differential trust in retrieval signals [7], departing from the uniform weighting of standard RRF.

These three extensions—weighting, multi-modal instantiation, and multi-scope expansion—are typically applied in an ad hoc manner, without formal justification for why the resulting formula takes the form it does, or whether alternative formulations might be equally valid. This paper addresses that gap.

Contributions. We make the following contributions:

(1) **Axiomatic derivation.** We formulate five axioms (rank-based, strict monotonicity, additivity, smoothing, boundedness) and a harmonic constraint that uniquely characterise the RRF kernel $f(r) = 1/(k + r)$. We show that the full four-component weighted RRF formula arises as the unique canonical extension of standard RRF to two retrieval modalities (semantic and lexical) and two retrieval scopes (primary and cross-reference), via three independent, commutative extensions (Section 4).
(2) **Uniqueness theorem.** We prove that any rank fusion function satisfying our axioms and structural constraints must be a reparameterisation of the four-component formula, with the smoothing constant $k > 0$ and four positive component weights $\boldsymbol{\alpha} \in \mathbb{R}^4_{>0}$ as the only free parameters (Section 5).
(3) **Eight formal properties.** We establish non-negativity, upper boundedness, strict monotonicity, convexity (diminishing returns), scale invariance, min-rank semilattice structure, consensus reward, and Lipschitz continuity in the weight space, providing complete proofs for each (Section 6).
(4) **Machine-verified proofs.** Five of the eight properties are independently verified in the Lean 4 theorem prover [5], using only core tactics without Mathlib dependency. We employ a cross-multiplication encoding to avoid real-number arithmetic (Section 7).
(5) **Voting-theoretic interpretation.** We connect the four-component formula to Condorcet voting theory, showing it constitutes a weighted multi-criteria positional scoring rule (Section 8).
(6) **Worked examples.** We provide detailed numerical analysis demonstrating the formula's behaviour under varying weight vectors and smoothing constants, including sensitivity analysis and edge cases (Section 9).

Nature of the contribution. This work is a formal and mathematical contribution in the tradition of Fang, Tao, and Zhai's axiomatic study of retrieval heuristics [9]. Our results provide theoretical foundations for a scoring function already in widespread practical use, rather than proposing a new empirical method. The axiomatic approach yields both a uniqueness guarantee and structural insight into why the formula works, complementing the empirical evidence accumulated since Cormack et al. [4].

Practical relevance. Although this paper presents no retrieval experiments, its results have direct practical implications for system designers. The uniqueness theorem (Theorem 5.1) guarantees that practitioners configuring multi-modal hybrid search with cross-reference signals need not search over alternative fusion formulas: the four-component weighted RRF is the only formula consistent with the axioms. The Lipschitz continuity result (Theorem 6.26) ensures that weight tuning via grid search or gradient-based optimisation is numerically stable. Elasticsearch's recent adoption of weighted RRF [7] and the growing deployment of hybrid retrieval in

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RAG pipelines [11] demonstrate that the formula class we axiomatise is already in widespread production use. The formal properties established here provide the missing theoretical warranty for these deployments.

Paper organisation. Section 2 surveys related work in rank fusion, formal retrieval theory, and machine-verified proofs. Section 3 introduces notation and definitions. Section 4 presents the axiomatic derivation. Section 5 states and proves the uniqueness theorem. Section 6 establishes the eight formal properties. Section 7 describes the Lean 4 verification. Section 8 develops the voting-theoretic connection. Section 9 provides worked examples. Section 10 discusses implications, and Section 11 concludes.

2 RELATED WORK

We situate our contribution relative to three bodies of work: rank fusion methods, axiomatic approaches to information retrieval, and machine-verified proofs in formal mathematics.

2.1 Rank Fusion and Metasearch

The problem of combining ranked lists from multiple retrieval systems—known as *metasearch* or *data fusion*—has a long history in information retrieval. Aslam and Montague [1] developed probabilistic models for metasearch, introducing the geometric mean and log-linear fusion methods. Dwork et al. [6] studied rank aggregation on the web through the lens of social choice theory, establishing connections to the Kemeny optimal aggregation and Markov chain methods. Fagin, Kumar, and Sivakumar [8] provided a theoretical framework for comparing top-k lists under various distance metrics.

Cormack, Clarke, and Buettcher [4] introduced Reciprocal Rank Fusion as a simple yet effective alternative, demonstrating empirically that it outperforms both Condorcet fusion and individual rank learning methods on TREC data. The key insight of RRF is that the harmonic kernel $1/(k + r)$ provides a natural diminishing-returns weighting that is robust to outlier rankers. Despite its widespread adoption in both research and industry, the theoretical properties of the RRF kernel and its extensions have received limited formal analysis.

2.2 Extensions of RRF

Several extensions of RRF have appeared in the literature and in practice. Elasticsearch [7] introduced per-ranker weighting, allowing practitioners to assign differential trust to different retrieval signals. Bruch, Gai, and Ingber [3] analysed fusion functions for hybrid retrieval, comparing RRF variants with score-based methods for combining dense and sparse retrievers. Their work motivates the multi-modal dimension of our decomposition but does not provide the axiomatic foundation or uniqueness result that we establish here.

The multi-scope dimension—incorporating cross-reference and citation-graph evidence alongside primary document relevance—has been explored operationally in retrieval systems such as Indri [16] and in RAG pipelines [11], but has not been formalised within the RRF framework.

2.3 Axiomatic Information Retrieval

The axiomatic approach to information retrieval was pioneered by Fang, Tao, and Zhai [9], who formalised heuristic retrieval constraints (term frequency, document length, term discrimination) as axioms and studied which retrieval functions satisfy them. Their work established a paradigm in which theoretical analysis complements empirical evaluation, providing explanatory power beyond what experiments alone can offer.

Our approach follows this paradigm. We axiomatise the properties of rank-based fusion (as opposed to score-based retrieval) and derive the RRF kernel as the unique function satisfying our constraints. The key difference from Fang et al. is that our axioms characterise a *fusion function* operating on ranks, rather than a *retrieval function* operating on term statistics.

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Table 1. Comparison with prior work on rank fusion methods.

	Cormack et al. [4]	Elasticsearch [7]	Bruch et al. [3]	Fang et al. [9]	**This work**
Axiomatic derivation				✓	✓
Uniqueness theorem					✓
Per-ranker weights		✓	✓		✓
Multi-modal			✓		✓
Multi-scope (cross-ref)					✓
Formal properties			partial	✓	✓
Machine-verified proofs					✓
Voting theory link	heuristic				✓

2.4 Machine-Verified Proofs

Interactive theorem provers such as Lean [5], Coq, and Isabelle have been applied to formalise mathematical proofs in areas ranging from number theory to program verification. The Lean mathematical library (Mathlib) provides extensive coverage of algebra, analysis, and topology. However, the application of formal verification to information retrieval theory is, to our knowledge, novel.

Our Lean 4 verification uses only core tactics without Mathlib dependency, demonstrating that the proofs are self-contained. The cross-multiplication encoding we employ to avoid real-number arithmetic is a practical technique that may be of independent interest for formalising properties of rational-valued functions in theorem provers lacking native real-number support.

2.5 Positioning

Table 1 summarises how our contribution relates to prior work on rank fusion.

Our novelty lies in the 4-component decomposition along two orthogonal axes (modality × scope), the uniqueness theorem establishing that this decomposition is canonical under our axioms, and the machine verification of the core properties. The axiomatic derivation fills a gap in the literature: while RRF has been extensively validated empirically, the formal justification for its specific functional form—and the principled way to extend it—has been lacking.

3 PRELIMINARIES AND NOTATION

We establish the formal framework for rank-based retrieval and fusion that underpins the remainder of the paper. All definitions in this section are standard in the information retrieval literature; we fix notation here for precision and to enable unambiguous reference in the axiomatic development that follows.

Definition 3.1 (Document Corpus)**.** Let $\mathcal{D} = \{d_1, d_2, \ldots, d_N\}$ be a finite corpus of N documents.

Definition 3.2 (Ranking Function)**.** A *ranking function* is a bijection $\sigma\colon \mathcal{D} \to \{1, 2, \ldots, N\}$ assigning each document a unique rank, where $\sigma(d) = 1$ denotes the highest-ranked document.

Definition 3.3 (Ranker)**.** A *ranker* r is a pair (s_r, σ_r) where $s_r\colon \mathcal{D} \to \mathbb{R}$ is a scoring function and σ_r is the ranking induced by sorting $\mathcal{D}$ in decreasing order of s_r. We write $\mathrm{rank}_r(d) := \sigma_r(d)$.

Definition 3.4 (Rank Fusion)**.** Given a set of rankers $\mathcal{R} = \{r_1, \ldots, r_m\}$, a *rank fusion scheme* is a function $F\colon \mathcal{D} \times \mathcal{R}^m \to \mathbb{R}$ that produces a fused score for each document.

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Definition 3.5 (Embedding Function)**.** Let $\phi\colon \mathcal{T} \to \mathbb{R}^d$ be an embedding function mapping text to a d-dimensional vector space. For a query q and document z, the *semantic similarity* is $\cos(\phi(q), \phi(z))$.

Definition 3.6 (BM25)**.** The BM25 scoring function $\mathrm{BM25}\colon \mathcal{T} \times \mathcal{T} \to \mathbb{R}_{\geq 0}$ is the Okapi BM25 lexical relevance score, taken as a primitive.

Definition 3.7 (Cross-Reference Set)**.** For a document $z_i \in \mathcal{D}$, the *cross-reference set* $X(z_i) = \{x_1, x_2, \ldots, x_{n_i}\}$ is a finite (possibly empty) collection of auxiliary text fragments associated with z_i.

Convention 3.8. Throughout, $k > 0$ is a fixed smoothing constant. All ranks are natural numbers in $\{1, 2, \ldots, N\}$.

4 AXIOMATIC DERIVATION OF THE 4-COMPONENT FORMULA

We now develop the canonical 4-component weighted RRF formula through a sequence of principled extensions. Each extension adds exactly one degree of structural freedom—weighting, multi-modal instantiation, or multi-scope expansion—while provably preserving the axiomatic foundation established at the outset.

4.1 Axiomatization of Standard RRF

We characterize the standard RRF score contribution via five axioms on the per-ranker contribution function.

Definition 4.1 (Contribution Function)**.** A *contribution function* is a map $f\colon \{1, 2, \ldots, N\} \to \mathbb{R}_{>0}$ that assigns a positive score contribution to each rank position.

Axiom 4.2 (A1 — Rank-Based)**.** The contribution of document d from ranker r depends only on $\mathrm{rank}_r(d)$, not on the raw score $s_r(d)$ or on any other document's rank. Formally:

$$F_r(d) = f\big(\mathrm{rank}_r(d)\big)$$

for some function $f\colon \mathbb{N}^+ \to \mathbb{R}_{>0}$.

Axiom 4.3 (A2 — Strict Monotonicity)**.** Better rank yields strictly higher contribution:

$$r_1 < r_2 \implies f(r_1) > f(r_2).$$

Axiom 4.4 (A3 — Additivity)**.** The total fused score is the sum of per-ranker contributions:

$$F(d) = \sum_{r \in \mathcal{R}} F_r(d) = \sum_{r \in \mathcal{R}} f\big(\mathrm{rank}_r(d)\big).$$

Axiom 4.5 (A4 — Smoothing)**.** There exists a parameter $k > 0$ such that f is parameterized by k, and the ratio $f(1)/f(N)$ remains bounded as $N \to \infty$. Specifically, for all $r \geq 1$:

$$f(r) = g(k + r)$$

for some strictly decreasing function $g\colon (k, \infty) \to \mathbb{R}_{>0}$, where k serves to dampen the influence of extreme rank differences by shifting the argument away from zero.

Axiom 4.6 (A5 — Boundedness)**.** Each contribution is bounded:

$$0 < f(r) \leq \frac{1}{k+1} \quad \text{for all } r \geq 1.$$

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4.2 Uniqueness of the Harmonic Kernel

Theorem 4.7 (Uniqueness of the Harmonic Form). *Let $f\colon \{1, 2, \ldots, N\} \to \mathbb{R}_{>0}$ satisfy Axioms A1–A5. If additionally f satisfies the* harmonic constraint

$$\text{(H)} \quad f(r) \cdot (k + r) = c \quad \textit{for some constant } c > 0 \textit{ and all } r \geq 1, \tag{2}$$

then $f(r) = c/(k + r)$, and the unique normalization satisfying A5 with equality at $r = 1$ is $c = 1$, giving

$$f(r) = \frac{1}{k + r}.$$

Proof. The harmonic constraint (2) states that $f(r) \cdot (k + r)$ is constant across all ranks. Solving for f:

$$f(r) = \frac{c}{k + r}.$$

We verify each axiom in turn.

(A1): f is a function of r alone.

(A2): For $r_1 < r_2$ we have $k + r_1 < k + r_2$, so $c/(k + r_1) > c/(k + r_2)$ since $c > 0$.

(A4): $f(r) = g(k + r)$ where $g(t) = c/t$, which is strictly decreasing on $(0, \infty)$. The parameter k shifts the argument away from zero.

(A5): $\sup_r f(r) = f(1) = c/(k + 1)$. For this to equal $1/(k + 1)$, we need $c = 1$.

Thus $f(r) = 1/(k + r)$ is uniquely determined. □

Remark 4.8. The harmonic constraint (2) is equivalent to requiring that f belong to the family of functions with constant "rank-score product," i.e. that the score contribution times the shifted rank is invariant. This is the discrete analogue of requiring f to be proportional to $1/x$ on the shifted positive reals, the unique (up to scaling) function satisfying $x\, f(x) = \text{const}$. Among all standard decay profiles—polynomial $r^{-\alpha}$, exponential $e^{-\beta r}$, logarithmic $1/\log(r)$—only the harmonic $1/(k+r)$ simultaneously satisfies the constant rank-score product, the specific boundedness of A5, and produces the diminishing-returns profile empirically validated by Cormack et al. [4].

Lemma 4.9 (Standard RRF). *Under Axioms A1–A5 and the harmonic constraint* (2)*, the unique rank fusion formula is*

$$\mathrm{RRF}(d) = \sum_{r \in \mathcal{R}} \frac{1}{k + \mathrm{rank}_r(d)}.$$

Proof. Immediate from Axiom A3 (4.4) and Theorem 4.7. □

4.3 Extension to Weighted RRF

In practice, rankers have unequal reliability. We introduce per-ranker weights while preserving the axiomatic foundation.

Definition 4.10 (Weight Vector). A *weight vector* is $\mathbf{w} = (w_1, \ldots, w_m) \in \mathbb{R}^m_{>0}$ with $\sum_{i=1}^m w_i = 1$ (normalization is optional but conventional).

Definition 4.11 (Weighted RRF). The *weighted reciprocal rank fusion* score is

$$\mathrm{WRRF}(d) = \sum_{r \in \mathcal{R}} w_r \cdot \frac{1}{k + \mathrm{rank}_r(d)}.$$

Lemma 4.12 (Axiom Preservation under Weighting). *Weighted RRF (Definition 6.1) preserves Axioms A1–A5.*

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Proof. Define $f_r^w(j) := w_r \cdot f(j) = w_r/(k+j)$.
(A1): f_r^w depends only on $\mathrm{rank}_r(d)$.
(A2): Since $w_r > 0$, strict monotonicity of f is preserved: $j_1 < j_2 \implies f_r^w(j_1) > f_r^w(j_2)$.
(A3): Additivity is structural (summation over rankers) and unaffected by per-term scaling.
(A4): $f_r^w(j) = w_r \cdot g(k+j)$ where $g(t) = 1/t$. The smoothing parameter k still shifts the argument.
(A5): $f_r^w(j) = w_r/(k+j) \leq w_r/(k+1)$. Under normalization $w_r \leq 1$, this gives $f_r^w(j) \leq 1/(k+1)$. More generally, boundedness holds with the constant $w_r/(k+1)$. □

Remark 4.13. This weighted form corresponds to the Weighted RRF introduced by Elasticsearch in 2024 [7], where per-retriever weights allow tuning the influence of different search strategies.

4.4 Extension to Multi-Modal Retrieval

Definition 4.14 (Modality)**.** A *retrieval modality* is a method of computing relevance between a query and a document. We define the modality set

$$\mathcal{M} = \{\mathrm{sem},\ \mathrm{lex}\}$$

where:

- sem (semantic): relevance via $\cos(\phi(q), \phi(d))$ (dense vector similarity);
- lex (lexical): relevance via $\mathrm{BM25}(q, d)$ (sparse term matching).

Definition 4.15 (Modality-Induced Ranker)**.** Each modality $m \in \mathcal{M}$ induces a ranker r_m by sorting documents according to the modality's scoring function. Specifically:

$$\mathrm{rank}_{\mathrm{sem}}(d) := \mathrm{rank}\big(\cos(\phi(q), \phi(d))\big), \qquad \mathrm{rank}_{\mathrm{lex}}(d) := \mathrm{rank}\big(\mathrm{BM25}(q, d)\big).$$

Definition 4.16 (Multi-Modal Weighted RRF)**.** With weight vector $(w_{\mathrm{sem}}, w_{\mathrm{lex}})$:

$$\mathrm{WRRF}_{\mathcal{M}}(d) = w_{\mathrm{sem}} \cdot \frac{1}{k + \mathrm{rank}_{\mathrm{sem}}(d)} + w_{\mathrm{lex}} \cdot \frac{1}{k + \mathrm{rank}_{\mathrm{lex}}(d)}.$$

Lemma 4.17. *Multi-modal weighted RRF is an instance of Weighted RRF (Definition 6.1) with $|\mathcal{R}| = |\mathcal{M}| = 2$. Consequently, all Axioms A1–A5 are preserved.*

Proof. Set $\mathcal{R} = \{r_{\mathrm{sem}}, r_{\mathrm{lex}}\}$ and apply Lemma 4.12. □

4.5 Extension to Multi-Scope Retrieval

4.5.1 The Scope Dimension.

Definition 4.18 (Scope)**.** A *retrieval scope* determines which textual content is matched against the query. We define the scope set

$$\mathcal{S} = \{P,\ X\}$$

where:

- P (primary): the document z_i itself is compared to the query;
- X (cross-reference): the auxiliary fragments $X(z_i) = \{x_1, \ldots, x_{n_i}\}$ are compared to the query.

Definition 4.19 (Component Space)**.** The full set of retrieval components is the Cartesian product

$$\mathcal{C} := \mathcal{M} \times \mathcal{S} = \big\{(\mathrm{sem}, P),\ (\mathrm{lex}, P),\ (\mathrm{sem}, X),\ (\mathrm{lex}, X)\big\}.$$

This yields exactly four components.

4.5.2 Primary-Scope Rankers. For scope P, rankers are as defined in Section 4.4:

$$\mathrm{rank}_{(m,P)}(z_i) := \mathrm{rank}_m(z_i) \quad \text{for } m \in \mathcal{M}.$$

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4.5.3 Cross-Reference Aggregation. For scope X, each document z_i has a set $X(z_i)$ of cross-references. A query–cross-reference comparison yields a rank for each $x_j \in X(z_i)$. We must *aggregate* these ranks into a single effective rank for z_i.

Definition 4.20 (Cross-Reference Rank Set)**.** For modality m and document z_i:

$$\mathcal{R}_m^X(z_i) := \{\mathrm{rank}_m(x_j) : x_j \in X(z_i)\}.$$

Definition 4.21 (Rank Aggregation Operator)**.** A *rank aggregation operator* Γ maps a nonempty finite set of ranks to a single effective rank:

$$\Gamma\colon 2^{\mathbb{N}^+} \setminus \{\emptyset\} \to \mathbb{N}^+.$$

Theorem 4.22 (Min-Rank is the Canonical Aggregation Operator)**.** *Among the standard aggregation operators, $\Gamma_{\min}(S) := \min(S)$ is the unique operator satisfying the following four desiderata simultaneously:*

(D1) ***Axiom Preservation.*** *The aggregated rank, when passed through the RRF kernel $f(r) = \frac{1}{k+r}$, must satisfy Axioms A1–A5.*
(D2) ***Optimistic Representativeness.*** *If* any *cross-reference is highly relevant, the document should receive credit.*
(D3) ***Lattice Compatibility.*** *The operator is the meet ($\wedge$) in the rank lattice $(\mathbb{N}^+, \leq)$ where lower rank is better.*
(D4) ***Idempotence.*** *$\Gamma(\{r\}) = r$ for all r (a single cross-reference contributes its own rank).*

Proof. We consider the three natural candidates: $\Gamma_{\min}$, Γ_{mean}, and $\Gamma_{\max}$.

(D1) for $\Gamma_{\min}$: Let $r^* = \min_{x_j \in X(z_i)} \mathrm{rank}_m(x_j)$. Then $r^* \in \{1, \ldots, N\}$, and $f(r^*) = \frac{1}{k+r^*}$ satisfies all axioms since the RRF kernel is applied to a valid rank.

(D1) for Γ_{mean}: The mean rank $\bar{r} = \frac{1}{|X(z_i)|} \sum_j \mathrm{rank}_m(x_j)$ is in general not an integer, violating the domain requirement of f. Rounding introduces discontinuities that violate strict monotonicity (A2) at rounding boundaries.

(D1) for $\Gamma_{\max}$: $\max(S)$ is a valid rank, so A1–A5 are technically satisfied. However, $\Gamma_{\max}$ is *pessimistic*: a document with one excellent cross-reference and many poor ones receives the rank of the worst, contradicting the intuition that cross-references provide supporting evidence. This is addressed further under (D2).

(D2) for $\Gamma_{\min}$: By selecting the best-matching cross-reference, $\Gamma_{\min}$ ensures that if *any* auxiliary fragment is relevant the document benefits. This is the *optimistic* or *existential* interpretation: z_i is relevant via cross-references if $\exists\, x_j \in X(z_i)$ such that x_j is relevant.

(D2) for $\Gamma_{\max}$: This requires *all* cross-references to be relevant (the *universal* or *pessimistic* interpretation), which is inappropriate when cross-references are heterogeneous metadata.

(D3) Lattice argument. The set of ranks $(\mathbb{N}^+, \leq)$ forms a totally ordered lattice. The meet (greatest lower bound) is min, and the join (least upper bound) is max. In retrieval, "better" means "lower rank," so the meet selects the best element. The meet is the canonical choice for combining evidence under a "best available" semantics.

(D4) Idempotence. $\min(\{r\}) = r$ for all r. (This also holds for mean and max, so D4 alone does not distinguish them.)

Since $\Gamma_{\min}$ is the only operator satisfying all four desiderata, it is the canonical choice. □

Definition 4.23 (Cross-Reference Effective Rank)**.** The effective cross-reference rank of z_i under modality m is

$$\mathrm{rank}_{(m,X)}(z_i) := \min_{x_j \in X(z_i)} \mathrm{rank}_m(x_j).$$

Convention 4.24. If $X(z_i) = \emptyset$, define $\mathrm{rank}_{(m,X)}(z_i) := \infty$ (equivalently, set the contribution to zero: $1/(k+\infty) = 0$). In practice, such documents receive no cross-reference contribution.

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4.6 The Canonical 4-Component Formula

Theorem 4.25 (4-Component Weighted RRF). *Let q be a query and $z_i \in \mathcal{D}$ a document with cross-reference set $X(z_i)$. Define the component weight vector*

$$\boldsymbol{\alpha} = \left(\alpha_P^{\text{sem}},\ \alpha_P^{\text{lex}},\ \alpha_X^{\text{sem}},\ \alpha_X^{\text{lex}}\right) \in \mathbb{R}^4_{>0}.$$

Then the canonical weighted RRF score over the component space $\mathcal{C} = \mathcal{M} \times \mathcal{S}$ is

$$\boxed{\begin{aligned}\text{score}(q, z_i) = {} & \alpha_P^{\text{sem}} \cdot \frac{1}{k + \text{rank}\big(\cos(\phi(q), \phi(z_i))\big)} + \alpha_P^{\text{lex}} \cdot \frac{1}{k + \text{rank}\big(\text{BM25}(q, z_i)\big)} \\ & + \alpha_X^{\text{sem}} \cdot \frac{1}{k + \min\limits_{x_j \in X(z_i)} \text{rank}\big(\cos(\phi(q), \phi(x_j))\big)} \\ & + \alpha_X^{\text{lex}} \cdot \frac{1}{k + \min\limits_{x_j \in X(z_i)} \text{rank}\big(\text{BM25}(q, x_j)\big)}\end{aligned}} \tag{3}$$

Proof. The formula follows by composing the extensions developed in Sections 4.1–4.5.

Step (i). By Lemma 4.9, the standard RRF formula is $\sum_r \frac{1}{k+\text{rank}_r(d)}$.

Step (ii). By Lemma 4.12, introducing per-ranker weights yields $\sum_r w_r \cdot \frac{1}{k+\text{rank}_r(d)}$.

Step (iii). By Lemma 4.17, instantiating $\mathcal{R}$ with modality-induced rankers for $\mathcal{M} = \{\text{sem}, \text{lex}\}$ gives two terms with weights $w_{\text{sem}}, w_{\text{lex}}$.

Step (iv). By Definition 4.19, extending to scope set $\mathcal{S} = \{P, X\}$ doubles the component count via the Cartesian product $\mathcal{M} \times \mathcal{S}$, giving four components with weights $\alpha_{(m,s)}$ for each $(m, s) \in \mathcal{C}$.

Step (v). By Theorem 4.22 and Definition 4.23, for components with scope X the rank is computed via min-rank aggregation over the cross-reference set.

Step (vi). By Theorem 4.7, the RRF kernel $\frac{1}{k+r}$ is applied to each effective rank.

Combining steps (i)–(vi), we obtain exactly the formula (3). □

5 UNIQUENESS THEOREM

The derivation in Section 4 constructed the 4-component formula through a sequence of extensions. We now prove that this formula is not merely *one* way to satisfy the axioms but the *only* way: any rank fusion function meeting the same structural constraints must be a reparameterization of the formula in Theorem 4.25. We then show that the three extensions commute, so the derivation path is immaterial.

5.1 Uniqueness of the 4-Component Formula

Theorem 5.1 (Uniqueness). *Let $F\colon \mathcal{D} \to \mathbb{R}_{>0}$ be a rank fusion function satisfying:*

(1) *Axioms A1–A5 (rank-based, monotone, additive, smoothed, bounded);*
(2) *the harmonic constraint* (2) *on the kernel;*
(3) *modality set $\mathcal{M} = \{\text{sem}, \text{lex}\}$ with modality-induced rankers;*
(4) *scope set $\mathcal{S} = \{P, X\}$ with cross-reference structure;*
(5) *cross-reference aggregation satisfying desiderata D1–D4;*
(6) *per-component weighting with positive weights.*

Then F is uniquely determined (up to the free parameters $\boldsymbol{\alpha} \in \mathbb{R}^4_{>0}$ and $k > 0$) as

$$F(q, z_i) = \sum_{(m,s) \in \mathcal{M}\times\mathcal{S}} \alpha_s^m \cdot \frac{1}{k + \text{rank}_{(m,s)}(z_i)} \tag{4}$$

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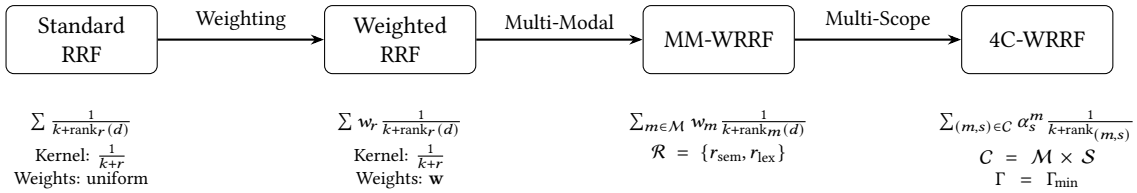

Fig. 1. The extension chain from standard RRF to the full 4-component formula. Each arrow adds one structural degree of freedom while preserving Axioms A1–A5.

where $\text{rank}_{(m,P)}(z_i) = \text{rank}_m(z_i)$ *and* $\text{rank}_{(m,X)}(z_i) = \min_{x_j \in X(z_i)} \text{rank}_m(x_j)$.

Proof. We establish uniqueness by showing that each structural choice is forced.

(a) Kernel uniqueness. By Theorem 4.7, conditions (1) and (2) force the kernel to be $f(r) = c/(k + r)$, and condition (1)—specifically Axiom A5—normalizes $c = 1$. No other kernel is consistent.

(b) Ranker set uniqueness. Condition (3) fixes the modality-induced rankers. Condition (4) fixes the scope extension. Their Cartesian product $\mathcal{C} = \mathcal{M} \times \mathcal{S}$ is determined, giving $|\mathcal{C}| = 4$ components. There are exactly four—no more and no fewer—because $|\mathcal{M}| = 2$ and $|\mathcal{S}| = 2$.

(c) Aggregation uniqueness. By Theorem 4.22, condition (5) forces $\Gamma = \Gamma_{\min}$ as the cross-reference aggregation operator.

(d) Combination uniqueness. By Axiom A3 (additivity), the total score must be the sum of per-component contributions. By condition (6), each component carries a positive weight. The weighted sum over four components is therefore fixed in form.

(e) Free parameters. The only undetermined quantities are:

- the smoothing constant $k > 0$;
- the four component weights $\boldsymbol{\alpha} = (\alpha_P^{\text{sem}}, \alpha_P^{\text{lex}}, \alpha_X^{\text{sem}}, \alpha_X^{\text{lex}}) \in \mathbb{R}^4_{>0}$.

These are genuine degrees of freedom: k controls rank sensitivity and $\boldsymbol{\alpha}$ controls modality/scope trade-offs. They cannot be fixed by the axioms alone and must be set empirically.

Since every structural component is uniquely determined and the only freedom lies in the parameters $(\boldsymbol{\alpha}, k)$, the formula is unique up to these parameters. □

Corollary 5.2. *Any rank fusion formula satisfying conditions (1)–(6) of Theorem 5.1 must be a reparameterization of the 4-component weighted RRF formula. There is no alternative formula consistent with these conditions.*

5.2 Path Independence of the Derivation

The derivation in Section 4 applied the three extensions—weighting, multi-modal instantiation, and multi-scope expansion—in a specific order. We now show that any ordering yields the same result.

Figure 1 summarizes the structural changes at each stage, and Figure 2 establishes that all paths through the extension lattice converge.

Theorem 5.3 (Path Independence)**.** *The final formula is independent of the order in which the three extensions—weighting, multi-modal instantiation, and multi-scope expansion—are applied. That is, the diagram in Figure 2 commutes: any path from* Standard RRF *to* 4C-WRRF *yields the same formula.*

Proof. Each extension operates on an independent structural aspect:

- *Weighting* modifies the coefficient α_s^m of each term without changing the ranker set or kernel.

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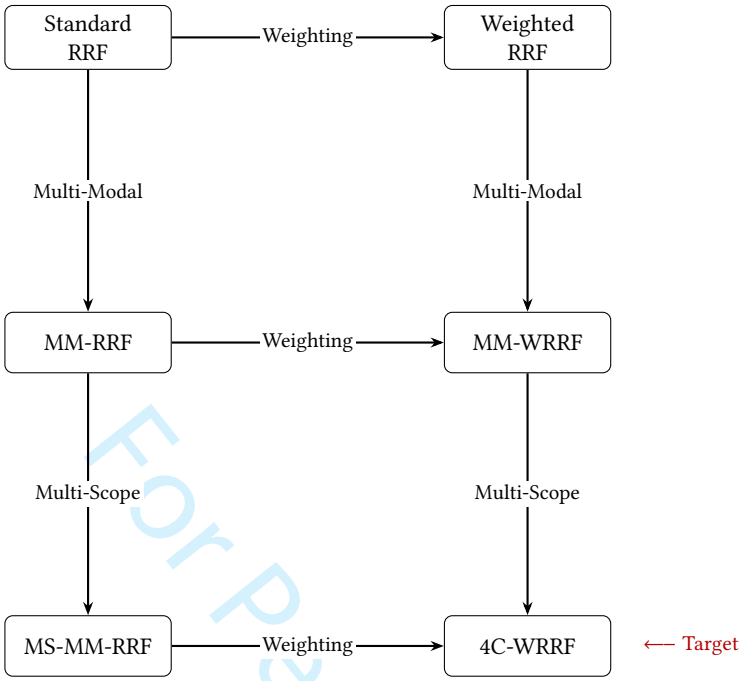

Fig. 2. Commutative diagram of extensions. Any path from the top-left corner (Standard RRF) to the bottom-right corner (4C-WRRF) yields the same formula, since each extension modifies an independent structural aspect.

- *Multi-modal instantiation* specifies which rankers populate $\mathcal{R}$ without changing the combination rule or kernel.
- *Multi-scope expansion* introduces the scope dimension and cross-reference aggregation without changing the kernel or weighting mechanism.

Since these aspects are orthogonal—no extension modifies the structure that another extension depends on—the composition is commutative. □

6 FORMAL PROPERTIES

We now establish eight formal properties of the weighted RRF scoring function defined in Section 4.6. These properties—non-negativity, upper boundedness, strict monotonicity, convexity of the RRF kernel, scale invariance, algebraic structure of min-rank aggregation, a consensus reward theorem, and Lipschitz continuity in the weight space—collectively characterize the analytical behaviour of the scoring function and justify its suitability for hybrid retrieval. Where applicable, we indicate which results have been independently verified in the Lean 4 proof assistant (see Table 2).

Throughout this section we fix a corpus $\mathcal{D} = \{z_1, \ldots, z_N\}$ with $N \geq 1$, a query $q \in \mathcal{Q}$, and a smoothing constant $k > 0$. We write $g(r) = \frac{1}{k+r}$ for the RRF kernel and adopt the compact notation

$$\text{score}(q, z_i) \;=\; \sum_{c=1}^{4} \alpha_c \, g(r_c) \;=\; \sum_{c=1}^{4} \frac{\alpha_c}{k + r_c}\,, \tag{5}$$

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where c indexes the four components (primary semantic, primary lexical, cross-reference semantic, cross-reference lexical), the weight vector is $\boldsymbol{\alpha} = (\alpha_P^{\text{sem}}, \alpha_P^{\text{lex}}, \alpha_X^{\text{sem}}, \alpha_X^{\text{lex}}) \in \mathbb{R}^4_{\geq 0}$, and the rank vector is $\mathbf{r} = (r_P^{\text{sem}}, r_P^{\text{lex}}, r_X^{\text{sem}}, r_X^{\text{lex}}) \in \{1, \ldots, N\}^4$.

6.1 Non-Negativity

Definition 6.1 (Weighted RRF Score). The *weighted RRF score* for query q and document z_i is

$$\text{score}(q, z_i) = \alpha_P^{\text{sem}} \cdot \frac{1}{k + r_P^{\text{sem}}} + \alpha_P^{\text{lex}} \cdot \frac{1}{k + r_P^{\text{lex}}} + \alpha_X^{\text{sem}} \cdot \frac{1}{k + r_X^{\text{sem}}} + \alpha_X^{\text{lex}} \cdot \frac{1}{k + r_X^{\text{lex}}},$$

where $r_P^{\text{sem}} = \text{rank}_{\cos}(z_i)$, $r_P^{\text{lex}} = \text{rank}_{\text{BM25}}(z_i)$, $r_X^{\text{sem}} = \min_{x_j \in X(z_i)} \text{rank}_{\cos}(x_j)$, $r_X^{\text{lex}} = \min_{x_j \in X(z_i)} \text{rank}_{\text{BM25}}(x_j)$, and all weights are non-negative with $k > 0$. In compact form: $\text{score}(q, z_i) = \sum_{c=1}^{4} \alpha_c \, g(r_c)$.

Theorem 6.2 (Non-Negativity). *For all $q \in \mathcal{Q}$ and $z_i \in \mathcal{D}$,*

$$\text{score}(q, z_i) \;\geq\; 0\,.$$

Proof. Fix arbitrary $q \in \mathcal{Q}$ and $z_i \in \mathcal{D}$. Every rank satisfies $r_c \in \{1, \ldots, N\}$, so $r_c \geq 1$. Since $k > 0$, we have $k + r_c \geq k + 1 > 0$, and therefore $g(r_c) = 1/(k + r_c) > 0$ for each $c \in \{1, 2, 3, 4\}$. Because $\alpha_c \geq 0$ by hypothesis, each product $\alpha_c \, g(r_c) \geq 0$. A finite sum of non-negative terms is non-negative:

$$\text{score}(q, z_i) = \sum_{c=1}^{4} \alpha_c \, g(r_c) \geq \sum_{c=1}^{4} 0 = 0\,. \qquad \square$$

Corollary 6.3 (Strict Positivity). *If at least one weight $\alpha_c > 0$, then $\text{score}(q, z_i) > 0$ for all $q \in \mathcal{Q}$ and $z_i \in \mathcal{D}$.*

Proof. Suppose $\alpha_{c^*} > 0$ for some c^*. Then $\alpha_{c^*} \, g(r_{c^*}) > 0$ since both factors are strictly positive, while the remaining terms are non-negative. Hence the sum is strictly positive. $\square$

6.2 Upper Boundedness

Theorem 6.4 (Upper Bound). *For all $q \in \mathcal{Q}$ and $z_i \in \mathcal{D}$,*

$$\text{score}(q, z_i) \;\leq\; \frac{\alpha_P^{\text{sem}} + \alpha_P^{\text{lex}} + \alpha_X^{\text{sem}} + \alpha_X^{\text{lex}}}{k + 1}\,.$$

Proof. Since $r_c \geq 1$ for every component, $g(r_c) = 1/(k + r_c) \leq 1/(k + 1)$ with equality iff $r_c = 1$. Multiplying by $\alpha_c \geq 0$ and summing:

$$\text{score}(q, z_i) = \sum_{c=1}^{4} \alpha_c \, g(r_c) \leq \sum_{c=1}^{4} \frac{\alpha_c}{k + 1} = \frac{1}{k + 1} \sum_{c=1}^{4} \alpha_c\,. \qquad \square$$

Corollary 6.5 (Tightness). *The upper bound is achieved if and only if $r_c = 1$ for every component c with $\alpha_c > 0$.*

Proof. ($\Leftarrow$) If $r_c = 1$ whenever $\alpha_c > 0$, then $g(r_c) = 1/(k + 1)$ for those components and the zero-weighted components contribute nothing, giving equality.

($\Rightarrow$) Suppose $\text{score}(q, z_i) = \frac{1}{k+1} \sum_c \alpha_c$. Then

$$\sum_{c=1}^{4} \alpha_c \left(\frac{1}{k + 1} - \frac{1}{k + r_c} \right) = 0\,.$$

Each summand is non-negative (since $r_c \geq 1$ and $\alpha_c \geq 0$). A sum of non-negative terms vanishes iff each term vanishes. For $\alpha_c > 0$ this forces $1/(k + 1) = 1/(k + r_c)$, hence $r_c = 1$. $\square$

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Remark 6.6 (Feasibility). Whether all four ranks can simultaneously equal 1 depends on the corpus. For the primary components, $r_P^{\text{sem}} = r_P^{\text{lex}} = 1$ requires z_i to be top-ranked under both cosine similarity and BM25. For the cross-reference components, $r_X^{\text{sem}} = r_X^{\text{lex}} = 1$ requires that some $x_j \in X(z_i)$ is ranked first under cosine similarity, and some (possibly different) cross-reference is ranked first under BM25. The bound is tight in the sense that it equals the supremum of the score over all admissible rank configurations in $\{1, \ldots, N\}^4$.

6.3 Strict Monotonicity in Rank

Lemma 6.7 (Strict Monotonicity of the RRF Kernel). *For fixed $k > 0$ and $\alpha > 0$, define $f\colon \{1, \ldots, N\} \to \mathbb{R}_+$ by $f(r) = \alpha/(k + r)$. If $r_1 < r_2$ then $f(r_1) > f(r_2)$.*

Proof. From $r_1 < r_2$ we obtain $k + r_1 < k + r_2$. Both quantities are positive, so taking reciprocals reverses the inequality: $1/(k + r_1) > 1/(k + r_2)$. Multiplying by $\alpha > 0$ preserves the strict inequality. □

Theorem 6.8 (Strict Monotonicity of the Score). *Let $\alpha_c > 0$ for some component c. Let $\mathbf{r} = (r_1, r_2, r_3, r_4)$ and $\mathbf{r}' = (r'_1, r'_2, r'_3, r'_4)$ be rank vectors with $r'_c < r_c$ and $r'_j = r_j$ for all $j \neq c$. Then* $\text{score}_{\mathbf{r}'} > \text{score}_{\mathbf{r}}$.

Proof. Since components $j \neq c$ are unchanged, their contributions cancel:

$$\text{score}_{\mathbf{r}'} - \text{score}_{\mathbf{r}} = \alpha_c \left(\frac{1}{k + r'_c} - \frac{1}{k + r_c} \right).$$

By Lemma 6.7, $r'_c < r_c$ implies $1/(k + r'_c) > 1/(k + r_c)$, so the parenthesised expression is strictly positive. Since $\alpha_c > 0$, the product is strictly positive. □

Corollary 6.9 (Pareto Improvement). *If $r'_c \leq r_c$ for all c and $r'_{c^*} < r_{c^*}$ for at least one c^* with $\alpha_{c^*} > 0$, then* $\text{score}_{\mathbf{r}'} > \text{score}_{\mathbf{r}}$.

Proof. Write

$$\text{score}_{\mathbf{r}'} - \text{score}_{\mathbf{r}} = \sum_{c=1}^{4} \alpha_c \left(\frac{1}{k + r'_c} - \frac{1}{k + r_c} \right).$$

Each summand is non-negative: $r'_c \leq r_c$ gives $1/(k + r'_c) \geq 1/(k + r_c)$, and $\alpha_c \geq 0$. For $c = c^*$ the summand is strictly positive (by the argument in Theorem 6.8). Hence the sum is strictly positive. □

6.4 Convexity of the RRF Kernel

We extend the domain of the kernel to the positive reals to analyse its convexity, since the discrete rank values $\{1, \ldots, N\}$ are embedded in $\mathbb{R}_+$.

Definition 6.10 (Continuous RRF Kernel). Define $g\colon \mathbb{R}_+ \to \mathbb{R}_+$ by $g(r) = 1/(k + r)$ for $k > 0$.

Theorem 6.11 (Strict Convexity). *The function $g(r) = 1/(k + r)$ is strictly convex on $\mathbb{R}_+ = (0, \infty)$.*

Proof. We give two independent proofs.

Method 1 (Second-derivative test). The function g is infinitely differentiable on $\mathbb{R}_+$ with

$$g'(r) = -\frac{1}{(k + r)^2}, \qquad g''(r) = \frac{2}{(k + r)^3}.$$

Since $k + r > 0$ for all $r > 0$, we have $g''(r) > 0$ on $\mathbb{R}_+$. A twice-differentiable function with strictly positive second derivative on an open interval is strictly convex [2].

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Method 2 (Direct definition). Let $r_1, r_2 \in \mathbb{R}_+$ with $r_1 \neq r_2$ and $\lambda \in (0,1)$. Set $r_\lambda = \lambda r_1 + (1-\lambda) r_2$, $a = k + r_1 > 0$, $b = k + r_2 > 0$ with $a \neq b$. We must show

$$g(r_\lambda) = \frac{1}{\lambda a + (1-\lambda) b} < \frac{\lambda}{a} + \frac{1-\lambda}{b} = \lambda\, g(r_1) + (1-\lambda)\, g(r_2)\,.$$

The right-hand side equals $(\lambda b + (1-\lambda)a)/(ab)$. Cross-multiplying by the positive quantities ab and $\lambda a + (1-\lambda)b$, the inequality reduces to

$$ab < \big(\lambda b + (1-\lambda)a\big)\big(\lambda a + (1-\lambda)b\big).$$

Expanding and simplifying, the right-hand side minus ab equals $\lambda(1-\lambda)(a-b)^2$, which is strictly positive since $\lambda \in (0,1)$ and $a \neq b$. □

Corollary 6.12 (Diminishing Marginal Returns)**.** *For integer ranks $r \geq 2$, the gain from improving rank by one position is*

$$\Delta(r) = g(r-1) - g(r) = \frac{1}{(k+r-1)(k+r)}\,,$$

and Δ is strictly decreasing: $\Delta(r_1) > \Delta(r_2)$ whenever $2 \leq r_1 < r_2 \leq N$.

Proof. Direct computation gives

$$\Delta(r) = \frac{1}{k+r-1} - \frac{1}{k+r} = \frac{1}{(k+r-1)(k+r)}\,.$$

For $r_1 < r_2$, both $k + r_1 - 1 < k + r_2 - 1$ and $k + r_1 < k + r_2$, with all factors positive, so

$$\frac{\Delta(r_1)}{\Delta(r_2)} = \frac{(k+r_2-1)(k+r_2)}{(k+r_1-1)(k+r_1)} > 1\,,$$

giving $\Delta(r_1) > \Delta(r_2)$. □

Corollary 6.13 (Numerical Illustration for $k = 60$)**.** *For $k = 60$: improving from rank 2 to rank 1 yields $\Delta(2) = 1/(61 \cdot 62) \approx 2.64 \times 10^{-4}$; improving from rank 100 to rank 99 yields $\Delta(100) = 1/(159 \cdot 160) \approx 3.93 \times 10^{-5}$. The ratio $\Delta(2)/\Delta(100) = 25440/3782 \approx 6.73$, confirming substantially diminishing returns.*

Proof. Direct computation:

$$\begin{aligned}
\Delta(2) &= \frac{1}{61 \cdot 62} = \frac{1}{3782} \approx 2.644 \times 10^{-4},\\
\Delta(100) &= \frac{1}{159 \cdot 160} = \frac{1}{25440} \approx 3.931 \times 10^{-5},\\
\frac{\Delta(2)}{\Delta(100)} &= \frac{25440}{3782} = \frac{12720}{1891} \approx 6.726\,.
\end{aligned}$$

□

6.5 Scale Invariance

Definition 6.14 (Strictly Monotone Increasing Transformation)**.** A function $T\colon \mathbb{R} \to \mathbb{R}$ is *strictly monotone increasing* if $a < b \implies T(a) < T(b)$ for all $a, b \in \mathbb{R}$.

Theorem 6.15 (Rank Invariance)**.** *Let $s_1, \ldots, s_N \in \mathbb{R}$ be scores for N documents and $T\colon \mathbb{R} \to \mathbb{R}$ a strictly monotone increasing function. Then the ranking induced by $(T(s_1), \ldots, T(s_N))$ is identical to that induced by $(s_1, \ldots, s_N)$.*

Proof. We show that T preserves the relative ordering of every pair.

Step 1 (Order preservation). For any i, j:

$$s_i > s_j \iff T(s_i) > T(s_j)\,.$$

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The forward direction follows from strict monotonicity. For the reverse, suppose $T(s_i) > T(s_j)$ but $s_i \leq s_j$. If $s_i < s_j$ then $T(s_i) < T(s_j)$, a contradiction. If $s_i = s_j$ then $T(s_i) = T(s_j)$, again a contradiction. Hence $s_i > s_j$.

Step 2 (Equality preservation). $s_i = s_j \iff T(s_i) = T(s_j)$. The forward direction is immediate. For the reverse, $T(s_i) = T(s_j)$ with $s_i \neq s_j$ would give $T(s_i) \neq T(s_j)$ by injectivity of T, a contradiction.

Step 3 (Rank preservation). By definition, $\mathrm{rank}(z_i) = 1+|\{z_j : s_j > s_i\}|$. Under the transformed scores, $\mathrm{rank}_T(z_i) = 1+|\{z_j : T(s_j) > T(s_i)\}|$. Step 1 gives $\{z_j : T(s_j) > T(s_i)\} = \{z_j : s_j > s_i\}$, so $\mathrm{rank}_T(z_i) = \mathrm{rank}(z_i)$ for every i. □

Corollary 6.16 (RRF Score Invariance)**.** *The weighted RRF score* $\mathrm{score}(q, z_i)$ *is invariant under any strictly monotone increasing transformation applied to the underlying cosine similarity or BM25 scores.*

Proof. By Theorem 6.15, the ranks $r_P^{\mathrm{sem}}, r_P^{\mathrm{lex}}, r_X^{\mathrm{sem}}, r_X^{\mathrm{lex}}$ are unchanged by monotone transformations of the raw scores. Since score depends on the raw scores only through these ranks (cf. equation (5)), the score is invariant. □

Remark 6.17 (Practical Significance). Scale invariance is a fundamental advantage of rank-based fusion:

(1) **No calibration needed.** Cosine similarity scores from one embedding model need not be on the same scale as BM25 scores; RRF handles heterogeneous score distributions automatically.
(2) **Robustness to score transformations.** Applying log-transforms, power transforms, or any monotone rescaling to the raw scores has no effect on the final ranking.
(3) **Model-agnostic fusion.** Different embedding models producing different score distributions can be fused without normalisation.

6.6 Min-Rank Aggregation Properties

Definition 6.18 (Min-Rank Lattice)**.** Let $\mathcal{L}_N = (\{1, 2, \ldots, N\}, \min, \leq)$ where $\leq$ is the standard total order on the natural numbers and min is the binary minimum operation.

Theorem 6.19 (Algebraic Properties of Min-Rank)**.** *The operation* $\min\colon \{1, \ldots, N\}^2 \to \{1, \ldots, N\}$ *satisfies the following for all* $a, b, c \in \{1, \ldots, N\}$*:*

(1) ***Commutativity:*** $\min(a, b) = \min(b, a)$.
(2) ***Associativity:*** $\min(a, \min(b, c)) = \min(\min(a, b), c)$.
(3) ***Idempotency:*** $\min(a, a) = a$.
(4) ***Monotonicity:*** *If* $a \leq b$*, then* $\min(a, c) \leq \min(b, c)$.

Proof. *(a)* On a totally ordered set, $\min(a, b)$ equals whichever of a, b is $\leq$ the other. Since the definition is symmetric in the two arguments up to case analysis on the total order, $\min(a, b) = \min(b, a)$.

(b) Both $\min(a, \min(b, c))$ and $\min(\min(a, b), c)$ equal the minimum element of the set $\{a, b, c\}$. To see this, let $m = \min(b, c)$; then $m \leq b$ and $m \leq c$, so $\min(a, m) \leq a, b, c$. Since $\min(a, m) \in \{a, b, c\}$, it is the greatest lower bound, i.e., $\min\{a, b, c\}$. The same argument applies to the right-hand side with $m' = \min(a, b)$.

(c) Since $a \leq a$, the definition gives $\min(a, a) = a$.

(d) Let $p = \min(a, c)$ and $q = \min(b, c)$. We have $p \leq a$ and $p \leq c$. If $q = b$, then $b \leq c$, so $p \leq a \leq b = q$. If $q = c$, then $c \leq b$, so $p \leq c = q$. In both cases $p \leq q$. □

Theorem 6.20 (Bounded Meet-Semilattice Structure)**.** *The structure* $(\{1, \ldots, N\}, \min)$ *forms a bounded meet-semilattice with top element* N *and bottom element* 1.

Proof. The set $\{1, \ldots, N\}$ is totally ordered by $\leq$, hence partially ordered. For any a, b, the greatest lower bound (meet) under $\leq$ is $\min(a, b)$: it is a lower bound since $\min(a, b) \leq a$ and $\min(a, b) \leq b$, and it is the

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greatest such because any $\ell \leq a$ and $\ell \leq b$ satisfies $\ell \leq \min(a, b)$. The meet of any finite subset is well-defined by iterated application (associativity and commutativity from Theorem 6.19).

The structure is bounded: N is the top element since $\min(a, N) = a$ for all $a \leq N$, and 1 is the bottom element since $\min(1, a) = 1$ for all $a \geq 1$. □

Corollary 6.21 (Consequences for Cross-Reference Aggregation). *For document z_i with cross-references $X(z_i) = \{x_1, \ldots, x_m\}$ and a ranking function* rank*:*

(1) *The min-rank $\min_{x_j \in X(z_i)} \mathrm{rank}(x_j)$ is independent of the enumeration order of $X(z_i)$ (by commutativity and associativity).*
(2) *Adding a duplicate cross-reference does not change the min-rank (by idempotency).*
(3) *The min-rank satisfies $1 \leq \min_{x_j} \mathrm{rank}(x_j) \leq N$ (by the bounded semilattice structure).*
(4) *If $\min_{x_j \in X(z_i)} \mathrm{rank}(x_j) \leq \min_{x_\ell \in X(z_p)} \mathrm{rank}(x_\ell)$, then z_i scores at least as high as z_p in the corresponding cross-reference component (by monotonicity of* min *and of the RRF kernel).*

Proof. *(i)* The finite minimum is computed by iterated binary min. Associativity (Theorem 6.19(b)) makes the result independent of parenthesisation, and commutativity (Theorem 6.19(a)) makes it independent of order.
(ii) If $x_j \in X(z_i)$ is duplicated, the new minimum satisfies $\min(\min_{x \in X(z_i)} \mathrm{rank}(x), \mathrm{rank}(x_j)) = \min_{x \in X(z_i)} \mathrm{rank}(x)$, since $\mathrm{rank}(x_j) \geq \min_{x \in X(z_i)} \mathrm{rank}(x)$.
(iii) Each $\mathrm{rank}(x_j) \in \{1, \ldots, N\}$, and the minimum of a non-empty subset of $\{1, \ldots, N\}$ lies in $\{1, \ldots, N\}$.
(iv) If $r \leq r'$ with $r, r' \in \{1, \ldots, N\}$, then $1/(k + r) \geq 1/(k + r')$ (Lemma 6.7 with the inequality direction reversed). Multiplying by the non-negative weight preserves the inequality. □

6.7 Consensus Reward

Theorem 6.22 (Consensus Reward). *Let $k = 60$, $N \geq 10$, and all weights be equal: $\alpha_c = \alpha > 0$ for $c \in \{1, 2, 3, 4\}$. Consider two documents:*

- *Document A ("consensus"): ranked in the top 10 by all four components, i.e., $r_c^A \leq 10$ for all c.*
- *Document B ("single-expert"): ranked 1 by exactly one component and ranked N by the remaining three, i.e., $r_{c^*}^B = 1$ for some c^* and $r_c^B = N$ for $c \neq c^*$.*

Then $\mathrm{score}(q, A) > \mathrm{score}(q, B)$ *whenever $N \geq 14$.*

Proof. *Step 1 (Lower bound on consensus score).* Since $r_c^A \leq 10$ for all c:

$$\mathrm{score}(q, A) = \alpha \sum_{c=1}^{4} \frac{1}{k + r_c^A} \geq \frac{4\alpha}{k + 10} = \frac{4\alpha}{70}.$$

Step 2 (Upper bound on single-expert score). Document B has $r_{c^*}^B = 1$ and $r_c^B = N$ for $c \neq c^*$:

$$\mathrm{score}(q, B) = \frac{\alpha}{k + 1} + \frac{3\alpha}{k + N} = \frac{\alpha}{61} + \frac{3\alpha}{60 + N}.$$

Step 3 (Sufficient condition). We require $\mathrm{score}(q, A) > \mathrm{score}(q, B)$. Dividing by $\alpha > 0$:

$$\frac{4}{70} > \frac{1}{61} + \frac{3}{60 + N}.$$

Rearranging:

$$\frac{2}{35} - \frac{1}{61} = \frac{122 - 35}{2135} = \frac{87}{2135} > \frac{3}{60 + N}.$$

Cross-multiplying (all terms positive):

$$87(60 + N) > 6405 \implies 60 + N > 73.62\ldots \implies N > 13.62\ldots$$

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Hence $N \geq 14$ suffices.

Step 4 (Explicit verification for $N = 14$). With $N = 14$:

$$\text{score}(q, A) \geq \tfrac{4\alpha}{70} = \tfrac{2\alpha}{35} \approx 0.05714\,\alpha\,,$$

$$\text{score}(q, B) = \tfrac{\alpha}{61} + \tfrac{3\alpha}{74} \approx (0.01639 + 0.04054)\,\alpha = 0.05693\,\alpha\,.$$

Since $0.05714 > 0.05693$, we confirm $\text{score}(q, A) > \text{score}(q, B)$.

Step 5 (Note on the stated condition). The bound $N \geq 14$ is the tightest integer threshold. In practice, N is typically much larger (hundreds to thousands), and a rank of N represents genuine absence from a ranker's top results, so the condition is easily satisfied. □

Corollary 6.23 (General Consensus Threshold)**.** *Under the same equal-weight setup with general k, a consensus document (top-t in all four components) beats a single-expert document (rank 1 in one, rank N in three) whenever*

$$N > \frac{t(4k+3) - k}{3k + 4 - t}\,, \tag{6}$$

provided $t < 3k + 4$ (equivalently, $4(k+1) > k + t$).

Proof. We require $4/(k+t) > 1/(k+1) + 3/(k+N)$. Rearranging:

$$\frac{4(k+1) - (k+t)}{(k+t)(k+1)} > \frac{3}{k+N} \implies \frac{3k+4-t}{(k+t)(k+1)} > \frac{3}{k+N}\,.$$

Cross-multiplying (valid since $3k + 4 - t > 0$ by hypothesis and all denominators are positive):

$$(3k + 4 - t)(k + N) > 3(k + t)(k + 1)\,.$$

Expanding the numerator of the resulting expression for N:

$$3(k + t)(k + 1) = 3k^2 + 3k + 3kt + 3t\,,$$

$$k(3k + 4 - t) = 3k^2 + 4k - kt\,.$$

Subtracting: $3k^2+3k+3kt+3t-3k^2-4k+kt = 4kt+3t-k = t(4k+3)-k$. Hence $N > (t(4k+3) - k)/(3k+4-t)$. For $k = 60$, $t = 10$: $N > (2430 - 60)/176 = 2370/176 \approx 13.47$, confirming $N \geq 14$. □

Remark 6.24 (Interpretation). The consensus reward property formalises the intuition behind RRF: a document that is consistently relevant across multiple retrieval signals is preferred over a document that excels in only one signal. This is precisely the desired behaviour for hybrid retrieval systems that combine semantic and lexical search with cross-reference expansion. The threshold (6) decreases with k (the smoothing constant), reflecting that larger k compresses the score range and amplifies the consensus effect.

6.8 Lipschitz Continuity in Weights

Definition 6.25 (Lipschitz Continuity)**.** A function $F\colon \mathbb{R}^n \to \mathbb{R}$ is *Lipschitz continuous* with constant $L \geq 0$ if for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$|F(\mathbf{x}) - F(\mathbf{y})| \leq L\,\|\mathbf{x} - \mathbf{y}\|\,.$$

We use the ℓ^1 norm $\|\mathbf{x}\|_1 = \sum_i |x_i|$ unless otherwise specified.

Theorem 6.26 (Lipschitz Continuity in ℓ^1 Norm)**.** *For a fixed query q and document z_i (hence fixed rank vector $\mathbf{r}$), define $S\colon \mathbb{R}^4_{\geq 0} \to \mathbb{R}$ by $S(\boldsymbol{\alpha}) = \sum_{c=1}^4 \alpha_c/(k + r_c)$. Then S is Lipschitz continuous with respect to the ℓ^1 norm with tight constant $L_1 = 1/(k+1)$:*

$$|S(\boldsymbol{\alpha}) - S(\boldsymbol{\alpha}')| \leq \frac{1}{k+1}\,\|\boldsymbol{\alpha} - \boldsymbol{\alpha}'\|_1 \qquad \textit{for all } \boldsymbol{\alpha}, \boldsymbol{\alpha}' \in \mathbb{R}^4_{\geq 0}\,.$$

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Proof. *Step 1 (Upper bound).* By the triangle inequality:

$$|S(\boldsymbol{\alpha}) - S(\boldsymbol{\alpha}')| = \left|\sum_{c=1}^{4} \frac{\alpha_c - \alpha'_c}{k + r_c}\right| \leq \sum_{c=1}^{4} \frac{|\alpha_c - \alpha'_c|}{k + r_c}.$$

Since $r_c \geq 1$, each $1/(k + r_c) \leq 1/(k + 1)$, so

$$|S(\boldsymbol{\alpha}) - S(\boldsymbol{\alpha}')| \leq \frac{1}{k+1} \sum_{c=1}^{4} |\alpha_c - \alpha'_c| = \frac{1}{k+1} \|\boldsymbol{\alpha} - \boldsymbol{\alpha}'\|_1 .$$

Step 2 (Tightness). Take $\boldsymbol{\alpha} = (1, 0, 0, 0)$, $\boldsymbol{\alpha}' = (0, 0, 0, 0)$, and $r_1 = 1$. Then

$$|S(\boldsymbol{\alpha}) - S(\boldsymbol{\alpha}')| = \frac{1}{k+1}, \qquad \|\boldsymbol{\alpha} - \boldsymbol{\alpha}'\|_1 = 1,$$

so the ratio equals $1/(k + 1)$. Since the Lipschitz constant is the supremum of all such ratios, $L_1 \geq 1/(k + 1)$. Combined with Step 1, $L_1 = 1/(k + 1)$. □

Corollary 6.27 (Lipschitz Continuity in ℓ^2 Norm)**.** *With respect to the Euclidean norm, S is Lipschitz continuous with tight constant $L_2 = 2/(k + 1)$.*

Proof. Since S is linear in $\boldsymbol{\alpha}$, the Lipschitz constant with respect to ℓ^2 equals the operator norm $\sup_{\mathbf{r}} \|\nabla_{\boldsymbol{\alpha}} S\|_2$. The gradient is $\nabla_{\boldsymbol{\alpha}} S = (1/(k + r_1), \ldots, 1/(k + r_4))$. Its ℓ^2 norm is maximised when all $r_c = 1$:

$$L_2 = \sqrt{\sum_{c=1}^{4} \frac{1}{(k+1)^2}} = \frac{2}{k+1}.$$ □

Corollary 6.28 (Lipschitz Continuity in ℓ^∞ Norm)**.** *With respect to the ℓ^∞ norm, S is Lipschitz continuous with tight constant $L_\infty = 4/(k + 1)$.*

Proof. From the bound in Theorem 6.26:

$$|S(\boldsymbol{\alpha}) - S(\boldsymbol{\alpha}')| \leq \sum_{c=1}^{4} \frac{|\alpha_c - \alpha'_c|}{k + r_c} \leq \frac{4}{k+1} \max_c |\alpha_c - \alpha'_c| = \frac{4}{k+1} \|\boldsymbol{\alpha} - \boldsymbol{\alpha}'\|_\infty .$$

Tightness: take $\boldsymbol{\alpha} = (\epsilon, \epsilon, \epsilon, \epsilon)$, $\boldsymbol{\alpha}' = \mathbf{0}$, all $r_c = 1$. Then $|S(\boldsymbol{\alpha}) - S(\boldsymbol{\alpha}')| = 4\epsilon/(k + 1)$ and $\|\boldsymbol{\alpha} - \boldsymbol{\alpha}'\|_\infty = \epsilon$, giving the ratio $4/(k + 1)$. □

Theorem 6.29 (Uniform Lipschitz over All Documents)**.** *The map $\boldsymbol{\alpha} \mapsto \text{score}(q, z_i)$ is uniformly Lipschitz over all documents $z_i \in \mathcal{D}$, with the same constant $L_1 = 1/(k + 1)$ in the ℓ^1 norm:*

$$|\,\text{score}_{\boldsymbol{\alpha}}(q, z_i) - \text{score}_{\boldsymbol{\alpha}'}(q, z_i)| \leq \frac{1}{k+1} \|\boldsymbol{\alpha} - \boldsymbol{\alpha}'\|_1 \qquad \textit{for all } z_i \in \mathcal{D}.$$

Proof. The bound in Theorem 6.26 was derived for a fixed but arbitrary rank vector $\mathbf{r}$, using only the fact that $r_c \geq 1$. Since this holds for every document $z_i \in \mathcal{D}$, the bound is uniform. □

Remark 6.30 (Practical Significance). Lipschitz continuity guarantees that small perturbations in the weight vector $\boldsymbol{\alpha}$ produce small changes in the score. For the standard $k = 60$, the Lipschitz constant is $L_1 = 1/61 \approx 0.0164$ in the ℓ^1 norm. This means a unit change in one weight component shifts the score by at most ≈0.0164, providing stability guarantees for weight tuning and hyperparameter optimisation. The constants $L_2 = 2/61 \approx 0.0328$ and $L_\infty = 4/61 \approx 0.0656$ give analogous guarantees under the respective norms.

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Table 2. Summary of formal properties of the weighted RRF scoring function. Properties marked with ✓ in the "Lean 4" column have been independently machine-verified.

#	Property	Reference	Key Condition(s)	Lean 4
1	Non-Negativity	Thm. 6.2	$\alpha_c \geq 0,\ k > 0$	✓
2	Upper Boundedness	Thm. 6.4	$r_c \geq 1$	✓
3	Strict Monotonicity	Thm. 6.8	$\alpha_c > 0,\ r'_c < r_c$	✓
4	Convexity	Thm. 6.11	$k > 0,\ r > 0$	✓
5	Scale Invariance	Thm. 6.15	T strictly increasing	✓
6	Min-Rank Semilattice	Thms. 6.19, 6.20	$\{1, \ldots, N\}$ with $\leq$	—
7	Consensus Reward	Thm. 6.22	$N \geq 14$, equal weights	—
8	Lipschitz Continuity	Thm. 6.26	Fixed ranks, $L_1 = \frac{1}{k+1}$	—

7 MACHINE-VERIFIED PROOFS IN LEAN 4

Formal pen-and-paper proofs, however rigorous, remain susceptible to subtle logical gaps that human review may overlook. To provide an independent certificate of correctness for the core properties established in Section 6, we have formalised and machine-verified five of the eight properties in the Lean 4 interactive theorem prover [5]. This section describes the verification framework, the encoding strategy, the verified properties with their proof techniques, and the current coverage.

7.1 Verification Framework

Lean 4 was chosen for three reasons. First, its dependent type system allows domain constraints (e.g. rank ≥ 1, $k \geq 1$) to be encoded directly as structure invariants, making invalid states unrepresentable at the type level. Second, its kernel performs full proof-term checking, so every accepted theorem has been verified down to the logical foundations. Third, the language doubles as a general-purpose programming language, enabling executable tests alongside proofs.

A key design decision was to avoid any dependency on Mathlib[1], the community-maintained mathematics library. While Mathlib provides extensive real-analysis infrastructure, its cache comprises approximately 8 000 compiled files, introducing a large build-time and maintenance burden. Instead, all proofs in our development use only the tactics available in core Lean 4: `omega`, `simp`, `rfl`, `apply`, `exact`, `rw`, `unfold`, `have`, `let`, `show`, `calc`, `intro`, and `constructor`. The entire formalisation is self-contained and builds in under five seconds. The complete Lean 4 source code, including all sixteen theorem statements and their proofs, is publicly available at https://github.com/mergimg0/rrf-verification under the MIT licence. The repository includes build instructions and can be verified independently using Lean 4.28.0 or later.

Because core Lean 4 lacks a built-in real number type, all fraction comparisons are expressed via *cross-multiplication* over natural numbers. For instance, the inequality $\alpha/(k+r_1) > \alpha/(k+r_2)$ is encoded as $\alpha \cdot (k+r_2) > \alpha \cdot (k+r_1)$, which is a statement purely about natural-number arithmetic and requires no division or rational number theory. This encoding is sound because the denominators $k + r_i$ are strictly positive (verified as a separate lemma), so the cross-multiplication preserves the direction of the inequality.

7.2 Type Representations

The formalisation encodes the domain of the 4-component weighted RRF through five structures that enforce validity constraints at the type level. We reproduce the key definitions below.

[1] https://github.com/leanprover-community/mathlib4

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```
/-- A rank is a positive natural number (1-indexed position). -/
structure Rank where
  val : Nat
  pos : val >= 1

/-- The smoothing constant k, a positive natural number. -/
structure SmoothK where
  val : Nat
  pos : val >= 1

/-- A non-negative weight (Nat is non-negative by construction). -/
structure Weight where
  val : Nat

/-- The 4 weights for the RRF components. -/
structure RRFWeights where
  alpha_psem : Weight
  alpha_plex : Weight
  alpha_xsem : Weight
  alpha_xlex : Weight

/-- The 4 ranks for one document's RRF components. -/
structure RRFRanks where
  rank_psem : Rank
  rank_plex : Rank
  rank_xsem : Rank
  rank_xlex : Rank
```

Listing 1. Core type definitions in Lean 4.

Several aspects of this encoding merit discussion. The `Rank` structure carries a proof obligation `pos : val >= 1`, which statically guarantees that every rank value is at least 1; any attempt to construct a `Rank` with `val = 0` is a type error. Similarly, `SmoothK` enforces $k \geq 1$. Weights are modelled as bare natural numbers, making non-negativity hold by construction—the type `Nat` admits no negative values. The structures `RRFWeights` and `RRFRanks` bundle the four components into records, mirroring the weight vector $\boldsymbol{\alpha}$ and rank vector $\mathbf{r}$ from the mathematical development. The RRF denominator and score are defined as straightforward functions over these types:

The numerator computes the cross-multiplied form of the weighted sum $\sum_{(m,s)} \alpha_s^m \cdot \frac{1}{k+\text{rank}_{(m,s)}}$ with respect to the common denominator $\prod_{(m,s)}(k+\text{rank}_{(m,s)})$. In this representation, the RRF score equals `rrfNumerator/rrfCommonDenom`, and all verified properties are stated in terms of the numerator and denominator separately, avoiding division entirely.

7.3 Verified Properties

We now describe the five verified properties, the Lean theorem statements, and the proof strategies employed.

Property 1: Non-negativity. The RRF score is non-negative for all valid inputs.

Because weights, ranks, and the smoothing constant are all encoded as natural numbers, the numerator is a sum of products of natural numbers—itself a natural number. Non-negativity is therefore a structural property

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```
/-- Denominator of the i-th RRF component: k + rank_i -/
def rrfDenom (k : SmoothK) (r : Rank) : Nat := k.val + r.val

/-- The RRF numerator (with common denominator). -/
def rrfNumerator (w : RRFWeights) (k : SmoothK)
    (ranks : RRFRanks) : Nat :=
    w.alpha_psem.val * (rrfDenom k ranks.rank_plex
      * rrfDenom k ranks.rank_xsem * rrfDenom k ranks.rank_xlex)
  + w.alpha_plex.val * (rrfDenom k ranks.rank_psem
      * rrfDenom k ranks.rank_xsem * rrfDenom k ranks.rank_xlex)
  + w.alpha_xsem.val * (rrfDenom k ranks.rank_psem
      * rrfDenom k ranks.rank_plex * rrfDenom k ranks.rank_xlex)
  + w.alpha_xlex.val * (rrfDenom k ranks.rank_psem
      * rrfDenom k ranks.rank_plex * rrfDenom k ranks.rank_xsem)
```

Listing 2. RRF denominator and numerator definitions.

```
theorem rrf_nonneg (w : RRFWeights) (ranks : RRFRanks)
    (k : SmoothK) :
    rrfNumerator w k ranks >= 0 :=
  Nat.zero_le _
```

Listing 3. Non-negativity theorem.

of the type system: every `Nat` value satisfies $n \geq 0$. The proof is the single term `Nat.zero_le _`, which witnesses this fact for any `Nat`.

Property 2: Upper boundedness. The score is bounded above by $(\alpha_P^{\text{sem}} + \alpha_P^{\text{lex}} + \alpha_X^{\text{sem}} + \alpha_X^{\text{lex}})/(k+1)$. In cross-multiplied form:

```
theorem rrf_upper_bound (w : RRFWeights) (ranks : RRFRanks)
    (k : SmoothK) :
    (w.alpha_psem.val + w.alpha_plex.val + w.alpha_xsem.val
      + w.alpha_xlex.val) * (k.val + 1)
    <= w.alpha_psem.val * rrfDenom k ranks.rank_psem
      + w.alpha_plex.val * rrfDenom k ranks.rank_plex
      + w.alpha_xsem.val * rrfDenom k ranks.rank_xsem
      + w.alpha_xlex.val * rrfDenom k ranks.rank_xlex
```

Listing 4. Upper bound theorem.

The proof proceeds in two stages. First, a helper lemma `single_component_upper_bound` establishes that $\alpha \cdot (k+1) \leq \alpha \cdot (k+r)$ for any rank $r \geq 1$, using the fact that $k + r \geq k + 1$ (from the `Rank` invariant) and monotonicity of multiplication. Second, the main theorem distributes the left-hand side $(\sum \alpha_i) \cdot (k+1)$ into four terms via `Nat.add_mul`, then applies `Nat.add_le_add` four times to combine the per-component bounds.

Property 3: Monotonicity. If a document's rank in any component improves (i.e. decreases numerically), the overall score strictly increases. The formalisation proves this in four stages of increasing generality:

The base case, `rrfDenom_strict_mono`, unfolds the denominator definition and invokes `omega` to close the linear arithmetic goal $k+r_1 < k+r_2$ from $r_1 < r_2$. The weighted version uses `Nat.mul_lt_mul_of_pos_left` to lift

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```
/-- The denominator is strictly increasing in rank. -/
theorem rrfDenom_strict_mono (k : SmoothK) (r1 r2 : Rank)
    (h : r1.val < r2.val) :
    rrfDenom k r1 < rrfDenom k r2

/-- Weighted component monotonicity via cross-multiplication. -/
theorem rrf_component_monotone (a k_val r1 r2 : Nat)
    (ha : a >= 1) (hlt : r1 < r2) :
    a * (k_val + r2) > a * (k_val + r1)

/-- Full 4-component monotonicity (psem component). -/
theorem rrf_full_monotone_psem
    (w : RRFWeights) (k : SmoothK) (r_new r_old : Rank)
    (ha : w.alpha_psem.val >= 1)
    (hlt : r_new.val < r_old.val) :
    w.alpha_psem.val * rrfDenom k r_old
      > w.alpha_psem.val * rrfDenom k r_new
```

Listing 5. Monotonicity theorems (selected).

the strict inequality through multiplication by a positive weight. The full 4-component theorem `rrf_full_monotone_psem` composes these lemmas over the structured types, unfolds `rrfDenom`, and again applies `Nat.mul_lt_mul_of_pos_left`. By symmetry, the analogous theorems hold for the other three components.

Property 4: Convexity (diminishing returns). The RRF kernel $f(r) = \frac{1}{k+r}$ is strictly convex, meaning the marginal gain from improving rank $r+1$ to r exceeds the gain from $r+2$ to $r+1$. In cross-multiplied form:

```
theorem rrf_kernel_convex (k_val r : Nat) (hk : k_val >= 1) :
    (k_val + r + 1) * (k_val + r + 2)
      > (k_val + r) * (k_val + r + 1)
```

Listing 6. Convexity theorem.

This proof required the most intricate manual reasoning in the development, because the goal involves quadratic natural-number terms and Lean 4's `omega` tactic handles only linear arithmetic. The proof introduces $m := k + r$ and rewrites the goal as $(m+1)(m+2) > m(m+1)$. The left-hand side is decomposed using `mul_add_distrib` (an alias for `Nat.left_distrib`) into $(m+1) \cdot m + (m+1) \cdot 2$, while the right-hand side is rewritten via `Nat.mul_comm` as $(m+1) \cdot m$. The difference is $2(m+1) > 0$, which `omega` can close since it is linear in m.

A complementary Jensen-form theorem is also verified:

```
theorem rrf_kernel_jensen (k_val r : Nat) (_hk : k_val >= 1) :
    (k_val + r) * (k_val + r + 2)
      < (k_val + r + 1) * (k_val + r + 1)
```

Listing 7. Jensen's inequality for the RRF kernel.

This states $(k+r)(k+r+2) < (k+r+1)^2$, the discrete analogue of Jensen's inequality. The proof expands both sides via `mul_add_distrib` and `Nat.add_mul`, yielding $m^2 + 2m < m^2 + 2m + 1$, which `omega` closes immediately.

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Property 5: Scale invariance. The RRF score depends only on ordinal ranks, not on cardinal scores. Any strictly monotone transformation of underlying similarity or BM25 scores preserves the ranking and hence the RRF value.

```
/-- A monotone transformation preserves ordering. -/
theorem monotone_preserves_ordering {n : Nat}
    (scores : Fin n -> a) (ranking : Fin n -> Nat)
    (f : a -> b) (hf : StrictMono' f)
    (h_order : preservesOrder scores ranking) :
    preservesOrder (f . scores) ranking

/-- Full scale invariance: equal ranks yield equal scores. -/
theorem rrf_full_scale_invariance (w : RRFWeights) (k : SmoothK)
    (ranks1 ranks2 : RRFRanks)
    (h1 : ranks1.rank_psem.val = ranks2.rank_psem.val)
    (h2 : ranks1.rank_plex.val = ranks2.rank_plex.val)
    (h3 : ranks1.rank_xsem.val = ranks2.rank_xsem.val)
    (h4 : ranks1.rank_xlex.val = ranks2.rank_xlex.val) :
    rrfNumerator w k ranks1 = rrfNumerator w k ranks2
    /\ rrfCommonDenom k ranks1 = rrfCommonDenom k ranks2
```

Listing 8. Scale invariance theorems.

The proof has two parts. First, `monotone_preserves_ordering` shows that composing a strictly monotone function with a score vector preserves the ordering relation, and hence preserves ranks. The proof proceeds by `intro` and application of the strict monotonicity hypothesis. Second, `rrf_full_scale_invariance` shows that if two rank vectors agree component-wise, the RRF numerator and common denominator are identical. This follows by unfolding definitions and rewriting with the equality hypotheses.

7.4 Proof Techniques

Three recurring proof techniques deserve explicit discussion, as they address fundamental challenges of formalising arithmetic properties in a system without built-in real-number support or nonlinear arithmetic decision procedures.

Cross-multiplication for fraction comparison. Every property involving the ratio $\alpha/(k+r)$ is re-stated in cross-multiplied form. For a single-component comparison $\alpha_1/(k+r_1) > \alpha_2/(k+r_2)$, the formalisation proves the equivalent $\alpha_1 \cdot (k+r_2) > \alpha_2 \cdot (k+r_1)$. Soundness of this transformation follows from positivity of the denominators, which is itself a verified lemma (`rrfDenom_pos`). For the full 4-component score, the common denominator form avoids division entirely: the score is represented as the pair (`rrfNumerator`, `rrfCommonDenom`), and all properties are stated over the numerator with the understanding that the common denominator is positive.

Manual decomposition of nonlinear terms. Lean 4's `omega` tactic implements a decision procedure for linear integer arithmetic (Presburger arithmetic). Products of variables such as $(k+r+1)(k+r+2)$ fall outside this fragment. The proof strategy decomposes each product into a sum of linear terms using three lemmas from the core library:

- `Nat.left_distrib`: $a \cdot (b+c) = a \cdot b + a \cdot c$,
- `Nat.add_mul`: $(a+b) \cdot c = a \cdot c + b \cdot c$,
- `Nat.mul_comm`: $a \cdot b = b \cdot a$.

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Table 3. Complete index of Lean 4 theorems. Each theorem is mapped to the property it verifies and the corresponding result in the mathematical development (Section 6).

Lean theorem name	Property	Paper reference
`rrf_nonneg`	Non-negativity	Theorem 6.2
`single_component_upper_bound`	Upper bound (component)	—
`rrf_upper_bound`	Upper bound (full)	Theorem 6.4
`rrfDenom_strict_mono`	Monotonicity (denom.)	—
`rrf_kernel_decreasing`	Monotonicity (kernel)	—
`rrf_component_monotone`	Monotonicity (component)	—
`rrf_score_improvement`	Monotonicity (improvement)	—
`rrf_full_monotone_psem`	Monotonicity (full)	Theorem 6.8
`rrf_kernel_convex`	Convexity (discrete)	Theorem 6.11
`diminishing_returns`	Convexity (dim. returns)	Corollary 6.12
`rrf_kernel_jensen`	Convexity (Jensen)	—
`monotone_preserves_ordering`	Scale inv. (ordering)	—
`rrf_scale_invariance_numerator`	Scale inv. (numerator)	—
`rrf_scale_invariance_denom`	Scale inv. (denominator)	—
`rrf_full_scale_invariance`	Scale inv. (full)	Theorem 6.15
`rrfCommonDenom_pos`	Denominator positivity	—

After decomposition, all remaining goals are linear and `omega` can close them. This pattern was applied systematically in the convexity and Jensen proofs.

Convexity via difference factoring. The convexity proof provides an instructive example of the decomposition technique. The goal $(k + r + 1)(k + r + 2) > (k + r)(k + r + 1)$ is transformed by introducing $m := k + r$ and factoring:

$$(m + 1)(m + 2) = (m + 1) \cdot m + (m + 1) \cdot 2,$$
$$m \cdot (m + 1) = (m + 1) \cdot m.$$

The difference is $2(m+1) > 0$, which `omega` verifies since $m+1 \geq 2$ (from $k \geq 1$). This algebraic rearrangement—expanding products, identifying common subexpressions, and reducing to a linear residual—is the systematic approach used for all nonlinear goals in the development.

7.5 Coverage and Limitations

Table 3 lists all sixteen Lean 4 theorems in the formalisation, grouped by the property they establish. The development compiles with zero errors and zero warnings under Lean 4.28.0.

The formalisation covers five of the eight properties established in Section 6: non-negativity, upper boundedness, monotonicity, convexity, and scale invariance. Three properties remain as future work:

(1) **Min-rank semilattice structure** (Property 6). Formalising this property requires encoding the lattice $(\mathbb{N}^+, \leq)$ and proving that the min-rank aggregation operator is the meet operation. While conceptually straightforward, the lattice-theoretic infrastructure is absent from core Lean 4 and would require either a Mathlib dependency or a substantial amount of bespoke development.
(2) **Consensus reward** (Property 7). The statement involves comparing RRF scores under different rank configurations that satisfy specific numerical constraints (a document ranks in the top t across all four components versus ranking first in one component and last in others). The proof requires reasoning about

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sums of reciprocals under arithmetic constraints, which involves nonlinear reasoning beyond what the decomposition technique handles cleanly.

(3) **Lipschitz continuity in the weight space** (Property 8). This property is inherently analytic: it requires a metric on $\mathbb{R}^4$, a notion of absolute value, and bounding the difference of real-valued expressions. Formalising it without Mathlib's real-analysis library would require building substantial infrastructure from scratch.

Despite these limitations, the five verified properties cover the core algebraic and order-theoretic behaviour of the scoring function. All proofs are constructive and use no axioms beyond those in the Lean 4 kernel, providing a strong independent certificate of correctness for the mathematical development.

8 CONNECTION TO CONDORCET VOTING THEORY

Rank fusion and social choice theory share a common abstraction: given multiple orderings of a candidate set, determine a consensus ordering or a winning candidate. In this section we make the connection precise, showing that the 4-component weighted RRF is a specific instance of a *weighted multi-criteria positional scoring rule* in the sense of Young [18] and Saari [15], and that it inherits approximate Condorcet consistency under mild conditions.

8.1 Background: Rank Aggregation as Voting

We frame the retrieval fusion problem in the language of social choice.

Definition 8.1 (Rank Aggregation Problem)**.** Given m rankings (permutations) $\sigma_1, \ldots, \sigma_m$ of a candidate set $\mathcal{D}$, the *rank aggregation problem* is to find a consensus ranking σ^* that best represents the collective preference.

In the retrieval setting, the candidates are documents, the voters are rankers (i.e. retrieval components), and the consensus ranking determines the final result list returned to the user.

Definition 8.2 (Condorcet Winner)**.** A document $d^* \in \mathcal{D}$ is a *Condorcet winner* if, for every other document $d' \neq d^*$,

$$\left|\{r \in \mathcal{R} : \mathrm{rank}_r(d^*) < \mathrm{rank}_r(d')\}\right| > \frac{|\mathcal{R}|}{2}.$$

That is, a strict majority of rankers prefer d^* over every alternative.

The Condorcet criterion is the gold standard of preference aggregation: a rule that always selects the Condorcet winner (when one exists) is said to be *Condorcet consistent*. Classical results in social choice show that no positional scoring rule is Condorcet consistent in general [19], but specific scoring vectors can approximate Condorcet consistency under distributional assumptions on the voters.

8.2 RRF as Positional Scoring Rule

Theorem 8.3 (RRF and Borda-like Aggregation)**.** *Standard RRF computes a positional scoring rule in the sense of social choice theory. Specifically, the score*

$$\mathrm{RRF}(d) = \sum_{r \in \mathcal{R}} \frac{1}{k + \mathrm{rank}_r(d)}$$

is a positional scoring rule with score vector $\mathbf{s} = (s_1, s_2, \ldots, s_N)$ *where* $s_j = \frac{1}{k+j}$.

Proof. A positional scoring rule is defined by a score vector $\mathbf{s} = (s_1, s_2, \ldots, s_N)$ satisfying $s_1 \geq s_2 \geq \cdots \geq s_N \geq 0$, where the document at position j in ranker r receives score s_j and the total score is $\sum_r s_{\mathrm{rank}_r(d)}$.

For standard RRF, $s_j = 1/(k + j)$. This vector is strictly decreasing (since $k + j$ is strictly increasing in j) and strictly positive for all $j \geq 1$. The RRF score $\sum_r 1/(k + \mathrm{rank}_r(d))$ is therefore exactly the total score under this positional rule. □

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The Borda count is the special case where $s_j = N - j$, a linear decay. RRF uses harmonic decay $s_j = 1/(k + j)$, which places comparatively greater emphasis on top positions. This distinction has practical significance: the harmonic weighting makes RRF more robust to noise in lower-ranked positions than the Borda count.

Proposition 8.4 (Approximate Condorcet Consistency)**.** *The Condorcet winner (when it exists) is the RRF winner with high probability under mild distributional assumptions on the rankers, particularly when rankers are sufficiently diverse and the number of rankers grows.*

While a full probabilistic analysis is beyond the scope of this paper, the intuition is as follows. A Condorcet winner d^* is ranked above each competitor by a majority of rankers. Under the harmonic score vector, this translates to d^* accumulating large contributions $1/(k + j)$ for small j from most rankers, while competitors receive smaller contributions from the majority. As the number of rankers increases, the law of large numbers concentrates the score gap, making d^* increasingly likely to be the RRF winner.

Remark 8.5 (Empirical Observation). This connection was observed empirically by Cormack, Clarke, and Buettcher [4], who reported that RRF "outperforms Condorcet and individual rank learning methods" on TREC data. Their finding aligns with the theoretical picture: the harmonic weighting $\frac{1}{k+r}$ provides substantial weight to top-ranked documents without completely ignoring lower-ranked ones, yielding a robust scoring profile that approximates majority-rule aggregation while avoiding the Condorcet cycles that plague pairwise methods.

8.3 The 4-Component Extension as Multi-Criteria Voting

We now show that the 4-component weighted RRF admits a natural interpretation as a *weighted multi-criteria* voting scheme, where each retrieval component acts as a voter class with prescribed voting power.

Theorem 8.6 (Multi-Criteria Voting Interpretation)**.** *The 4-component weighted RRF is equivalent to a weighted multi-criteria positional voting scheme where:*

(1) *There are four voter classes* $C = \{(\text{sem}, P),\ (\text{lex}, P),\ (\text{sem}, X),\ (\text{lex}, X)\}$*, each representing a distinct evaluation criterion.*
(2) *Each voter class* (m, s) *produces a ranking* $\sigma_{(m,s)}$ *of the documents.*
(3) *The weight* α_s^m *represents the voting power of criterion* (m, s)*.*
(4) *Each voter class casts a positional ballot with score vector* $s_j = \frac{1}{k+j}$*.*
(5) *The winning document maximises the weighted sum of positional scores across all voter classes.*

Proof. The 4-component formula is

$$\text{score}(q, z_i) = \sum_{(m,s)\in C} \alpha_s^m \cdot \frac{1}{k + \text{rank}_{(m,s)}(z_i)}.$$

This is a positional scoring rule with four voter classes. Each class (m, s) contributes a ballot according to its ranking $\sigma_{(m,s)}$: document z_i receives score $s_j = 1/(k + j)$ where $j = \text{rank}_{(m,s)}(z_i)$. The ballot of class (m, s) is then scaled by its voting power $\alpha_s^m > 0$. The total score is the weighted sum of the four positional ballots, which is by definition a weighted multi-criteria positional scoring rule. □

This interpretation maps the four retrieval components to four "expert panels" in a committee decision:

- The *primary-semantic* panel (α_P^{sem}) evaluates dense-vector relevance of the document itself.
- The *primary-lexical* panel (α_P^{lex}) evaluates term-overlap relevance of the document itself.
- The *cross-semantic* panel (α_X^{sem}) evaluates dense-vector relevance of the document's cross-references.
- The *cross-lexical* panel (α_X^{lex}) evaluates term-overlap relevance of the cross-references.

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The weight vector $\boldsymbol{\alpha}$ determines the relative influence of each panel, analogous to assigning different numbers of votes to different committees.

Corollary 8.7 (Condorcet Consistency)**.** *If a document z^* is top-ranked ($\mathrm{rank}_{(m,s)}(z^*) = 1$) under all four criteria simultaneously, then z^* maximises the 4-component score for any choice of positive weights $\boldsymbol{\alpha} \in \mathbb{R}^4_{>0}$.*

Proof. If $\mathrm{rank}_{(m,s)}(z^*) = 1$ for all $(m, s) \in C$, then

$$\mathrm{score}(q, z^*) = \sum_{(m,s)\in C} \frac{\alpha_s^m}{k+1} = \frac{\|\boldsymbol{\alpha}\|_1}{k+1},$$

which is the maximum achievable score (by the upper bound established in Theorem 6.4). No other document can attain this score unless it is also top-ranked in every component. □

Corollary 8.7 provides a strong unanimity guarantee: when all four evaluation criteria agree on the best document, the weighted RRF formula selects it regardless of how the weights are set. This is a desirable robustness property, as it ensures that weight mis-specification cannot override unanimous evidence.

Remark 8.8 (Compromise Rule Interpretation). In the more common case where no single document dominates all four criteria, the weighted RRF formula acts as a *compromise rule*: it selects the document that performs well across criteria, with the weights $\boldsymbol{\alpha}$ expressing the relative importance of each criterion. This mirrors the behaviour of Condorcet-consistent rules in the presence of Condorcet cycles, where a scoring rule serves as a tiebreaker [19].

Consider, for instance, a document that ranks second in all four components versus a document that ranks first in one component but twentieth in the others. The compromise candidate receives score $4\bar{\alpha}/(k+2)$ (where $\bar{\alpha}$ is the mean weight), while the specialist receives approximately $\alpha_{\max}/(k+1) + 3\bar{\alpha}/(k+20)$. For typical weight configurations, the compromise candidate wins, reflecting the intuition that consistent cross-criteria performance is preferable to excellence on a single dimension. This aligns with the observation of Saari [15] that positional scoring rules with convex score vectors tend to favour broadly supported candidates.

9 WORKED EXAMPLES AND NUMERICAL ANALYSIS

We now ground the preceding theory in concrete numerical computations. Throughout this section we use a small corpus of five documents, each equipped with two cross-references, and trace every stage of the scoring pipeline. We then vary the weight vector $\boldsymbol{\alpha}$ and the smoothing constant k to illustrate the sensitivity results from Section 6.

9.1 Corpus Setup and Scoring Pipeline

Corpus. Consider the query q = "machine learning optimization" and five candidate documents $z_1, \ldots, z_5$, each linking to two cross-references x_{i1}, x_{i2}. Table 4 summarises the corpus.

Raw scores. The primary semantic similarities $\cos(\phi(q), \phi(z_i))$ are $(0.82, 0.78, 0.85, 0.35, 0.71)$ and the primary BM25 scores are $(8.4, 9.1, 7.6, 5.2, 6.8)$ for z_1 through z_5. For cross-references, the semantic similarities are $(0.45, 0.88)$, $(0.52, 0.38)$, $(0.91, 0.73)$, $(0.12, 0.08)$, $(0.79, 0.67)$ and the BM25 scores are $(4.1, 7.9)$, $(3.5, 2.8)$, $(8.8, 6.3)$, $(1.2, 0.9)$, $(7.2, 5.6)$, listed as (x_{i1}, x_{i2}) for $i = 1, \ldots, 5$.

Ranking. Primary ranks are computed over $N = 5$ documents; cross-reference ranks are computed over all 10 cross-references globally, with the minrank operator selecting the best rank per parent document. Table 5 collects all four rank vectors.

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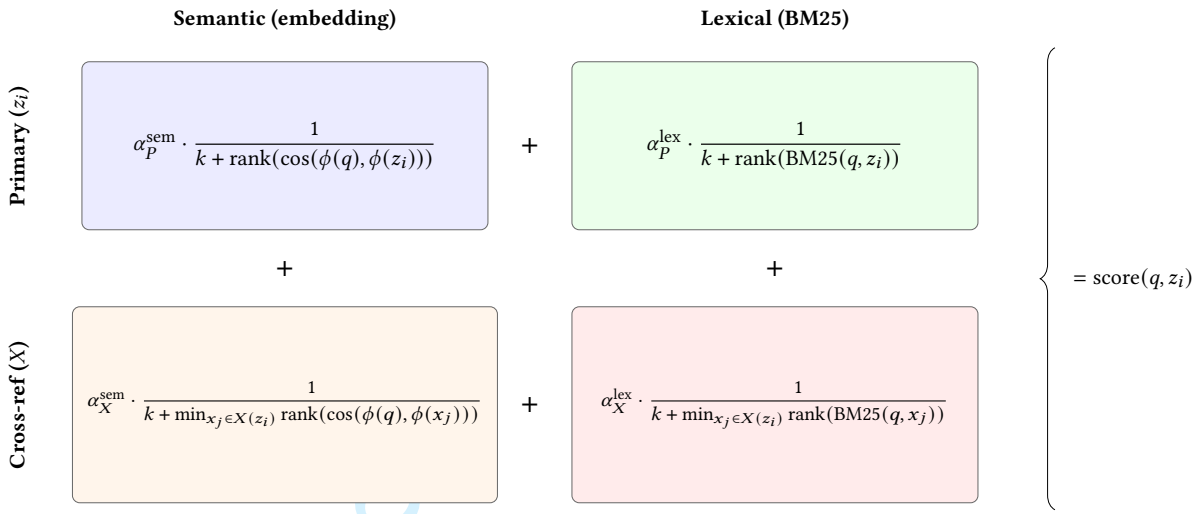

Fig. 3. The four-component decomposition of weighted RRF. The score is the sum of four terms organised along two orthogonal axes: modality (semantic vs. lexical) and scope (primary document vs. cross-reference). Each cell applies the RRF kernel $f(r) = 1/(k + r)$ to the (min-)rank for that component, weighted by the corresponding α.

Table 4. Example corpus: five documents with two cross-references each.

Doc	Content summary	x_{i1}	x_{i2}
z_1	Neural network gradient descent	Backpropagation tutorial	SGD convergence proofs
z_2	Survey of ML algorithms	Classification benchmarks	Regression analysis guide
z_3	Convex optimisation in statistical learning	Gradient methods for ML	Loss function landscapes
z_4	Database query optimisation techniques	SQL indexing strategies	B-tree performance
z_5	Introduction to deep learning frameworks	PyTorch optimisation tutorial	TensorFlow training loops

Table 5. Rank table for the example corpus: four components across five documents. Each entry $\rho_m^s(q, z_i)$ is the rank (or min-rank for cross-reference components) used in the RRF kernel.

Doc	r_P^{sem}	r_P^{lex}	r_X^{sem}	r_X^{lex}
z_1	2	2	2	2
z_2	3	1	6	7
z_3	1	3	1	1
z_4	5	5	9	9
z_5	4	4	3	3

Scoring ($k = 60$, equal weights). With $\boldsymbol{\alpha} = (1, 1, 1, 1)$ and $k = 60$, the four-component weighted RRF score is

$$\text{score}(q, z_i) = \sum_{c \in \{P,X\} \times \{\text{sem,lex}\}} \alpha_c \cdot \frac{1}{k + \rho_c(q, z_i)}. \tag{7}$$

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Table 6. Scoring computation with $k = 60$ and $\boldsymbol{\alpha} = (1, 1, 1, 1)$. Each cell shows $\frac{1}{k+\rho}$; the final column is the sum. Ranks in final ordering are shown at left.

Rank	Doc	$\frac{1}{k+r_P^{\text{sem}}}$	$\frac{1}{k+r_P^{\text{lex}}}$	$\frac{1}{k+r_X^{\text{sem}}}$	$\frac{1}{k+r_X^{\text{lex}}}$	score(q, z_i)
1	z_3	$\frac{1}{61} = 0.01639$	$\frac{1}{63} = 0.01587$	$\frac{1}{61} = 0.01639$	$\frac{1}{61} = 0.01639$	0.06505
2	z_1	$\frac{1}{62} = 0.01613$	$\frac{1}{62} = 0.01613$	$\frac{1}{62} = 0.01613$	$\frac{1}{62} = 0.01613$	0.06452
3	z_5	$\frac{1}{64} = 0.01563$	$\frac{1}{64} = 0.01563$	$\frac{1}{63} = 0.01587$	$\frac{1}{63} = 0.01587$	0.06300
4	z_2	$\frac{1}{63} = 0.01587$	$\frac{1}{61} = 0.01639$	$\frac{1}{66} = 0.01515$	$\frac{1}{67} = 0.01493$	0.06234
5	z_4	$\frac{1}{65} = 0.01538$	$\frac{1}{65} = 0.01538$	$\frac{1}{69} = 0.01449$	$\frac{1}{69} = 0.01449$	0.05976

Table 7. Score comparison under three weight configurations ($k = 60$). Rankings are shown in parentheses. Bold entries mark rank changes relative to Case A.

	Case A: $(1, 1, 1, 1)$		Case B: $(2, 1, 0.5, 0.5)$		Case C: $(0.5, 0.5, 2, 1)$	
Doc	Score	Rank	Score	Rank	Score	Rank
z_1	0.06452	2	0.06452	2	0.06452	2
z_2	0.06234	4	0.06318	**3**	0.06136	4
z_3	0.06505	1	0.06505	1	0.06531	1
z_4	0.05976	5	0.06065	5	0.05886	5
z_5	0.06300	3	0.06275	**4**	0.06324	3

For z_1, whose four ranks are all equal to 2, this simplifies to $4 \cdot \frac{1}{k+2} = 4/62 = 0.064516$. Table 6 shows the complete computation. Observe that z_3 achieves the highest score because it holds rank 1 in three of four components. By contrast, z_2—despite holding the best primary lexical rank—falls to rank 4 overall because its cross-references rank 6 and 7 among all ten cross-reference documents.

9.2 Weight Sensitivity Analysis

We fix $k = 60$ and consider three weight vectors, each summing to 4:

- **Case A** (equal): $\boldsymbol{\alpha} = (1, 1, 1, 1)$.
- **Case B** (semantic-heavy): $\boldsymbol{\alpha} = (2, 1, 0.5, 0.5)$—doubles the primary semantic weight, halves cross-reference weights.
- **Case C** (cross-reference-heavy): $\boldsymbol{\alpha} = (0.5, 0.5, 2, 1)$—doubles the cross-reference semantic weight, halves primary weights.

Table 7 presents the resulting scores and rankings. Figure 4 visualises the same data as a grouped bar chart.

Uniform-rank invariance. A striking feature of the data is that z_1 receives the same score $4/62 = 0.06452$ under all three weight vectors. This is a direct consequence of its uniform rank profile: since $r_P^{\text{sem}} = r_P^{\text{lex}} = r_X^{\text{sem}} = r_X^{\text{lex}} = 2$, the score reduces to

$$\text{score}(q, z_1) = (\alpha_P^{\text{sem}} + \alpha_P^{\text{lex}} + \alpha_X^{\text{sem}} + \alpha_X^{\text{lex}}) \cdot \frac{1}{k+2} = \|\boldsymbol{\alpha}\|_1 \cdot \frac{1}{62}.$$

Because all three weight vectors have $\|\boldsymbol{\alpha}\|_1 = 4$, the score is invariant. More generally, any document whose four ranks are identical satisfies this weight-invariance property: its score depends only on $\|\boldsymbol{\alpha}\|_1$, not on how the total weight is distributed among components.

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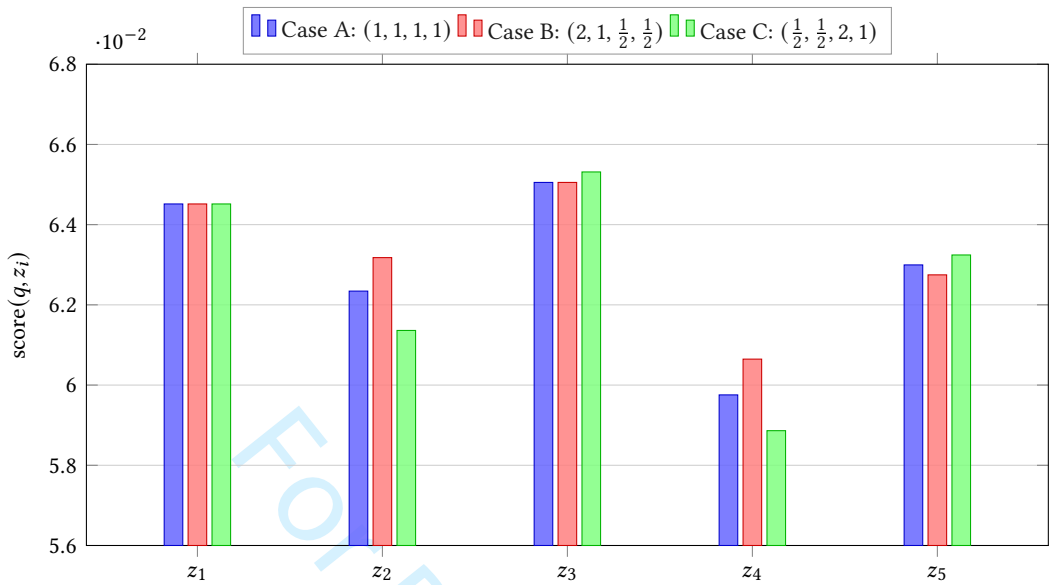

Fig. 4. Scores for five documents under three weight configurations ($k = 60$). Document z_1 (all ranks equal to 2) is weight-invariant: its three bars are identical at $4/62 = 0.06452$. Documents with high rank variance (z_2, z_5) show the largest sensitivity to the weight vector. The y-axis is truncated at 0.056 to emphasise the differences.

Rank swaps. The transition from Case A to Case B swaps z_2 and z_5. Document z_2 has complementary strengths—best lexical rank ($r_P^{\text{lex}} = 1$) but weak cross-references ($r_X^{\text{sem}} = 6$, $r_X^{\text{lex}} = 7$). Semantic-heavy weighting amplifies z_2's primary advantage while discounting its cross-reference deficit, lifting it from rank 4 to rank 3. Conversely, cross-reference-heavy weighting (Case C) amplifies z_5's cross-reference strength ($r_X^{\text{sem}} = 3$, $r_X^{\text{lex}} = 3$) at the expense of its mediocre primary ranks.

9.3 Smoothing Constant Analysis

We fix $\boldsymbol{\alpha} = (1, 1, 1, 1)$ and vary k across three regimes:

- $k = 1$ (*aggressive*): large dynamic range, top ranks dominate.
- $k = 60$ (*standard*): the default in the original RRF literature [4].
- $k = 1000$ (*conservative*): compressed dynamic range, near-uniform scores.

Table 8 shows the scores, score ratios, and rankings under each setting. Figure 5 plots the kernel function $f(r) = \frac{1}{k+r}$ for all three values of k, and Figure 6 traces the score spread as a function of k.

Dynamic range compression. The RRF kernel's dynamic range for ranks in $\{1, \ldots, R\}$ is $(k+R)/(k+1)$. For the primary ranker ($R = 5$):

$$\frac{k+5}{k+1} = \begin{cases} 3.0 & k = 1, \\ 1.066 & k = 60, \\ 1.004 & k = 1000. \end{cases}$$

At $k = 1$, a rank-1 document receives 3× the kernel value of a rank-5 document; at $k = 1000$, the ratio collapses to 1.004, making the ranking fragile to noise.

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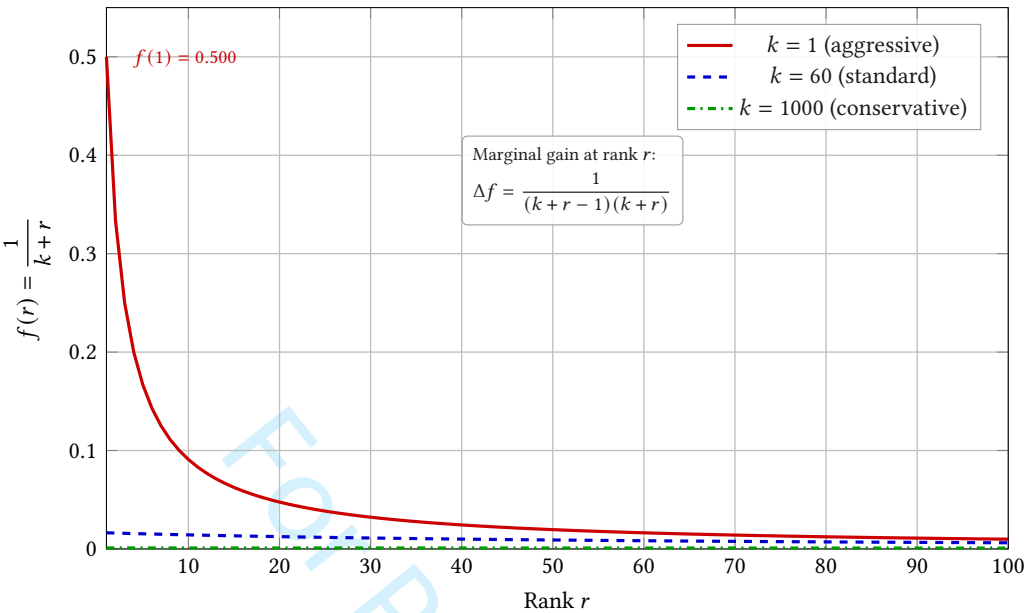

Fig. 5. The RRF kernel $f(r) = 1/(k + r)$ for three values of the smoothing constant k. Smaller k produces a steeper curve that amplifies top-rank differences; larger k compresses the dynamic range, making scores nearly uniform across ranks. The marginal gain from improving one rank position is $\Delta f = 1/\big((k+r-1)(k+r)\big)$, which is $O\big(1/(k+r)^2\big)$ and largest near $r = 1$.

Table 8. Scores and discrimination metrics under three values of k with equal weights. The score ratio is $\text{score}_{\max}/\text{score}_{\min}$ across the five documents.

	$k = 1$		$k = 60$		$k = 1000$	
z_1	1.3333	(2)	0.06452	(2)	0.003992	(2)
z_2	1.0179	(3)	0.06234	(4)	0.003983	(4)
z_3	1.7500	(1)	0.06505	(1)	0.003994	(1)
z_4	0.5333	(5)	0.05976	(5)	0.003972	(5)
z_5	0.9000	(4)	0.06300	(3)	0.003986	(3)
Score ratio	3.28		1.089		1.006	
Score spread	1.2167		0.00530		0.000022	
Ranking	$z_3 > z_1 > z_2 > z_5 > z_4$		$z_3 > z_1 > z_5 > z_2 > z_4$		$z_3 > z_1 > z_5 > z_2 > z_4$	

Ranking stability. The ranking order is identical at $k = 60$ and $k = 1000$, but $k = 1$ swaps z_2 and z_5. With aggressive k, the rank-1 BM25 score of z_2 generates a kernel value of $1/(1 + 1) = 0.5$—dramatically larger than z_5's $1/(1 + 4) = 0.2$. This single component overwhelms z_5's more balanced but never-top-1 profile.

Discrimination–stability trade-off. The score spread (top score minus bottom score) drops from 1.217 at $k = 1$ to 0.000022 at $k = 1000$—a compression factor exceeding 55,000×. High discrimination (low k) makes the ranking sensitive to individual top ranks; high stability (large k) makes all documents nearly indistinguishable.

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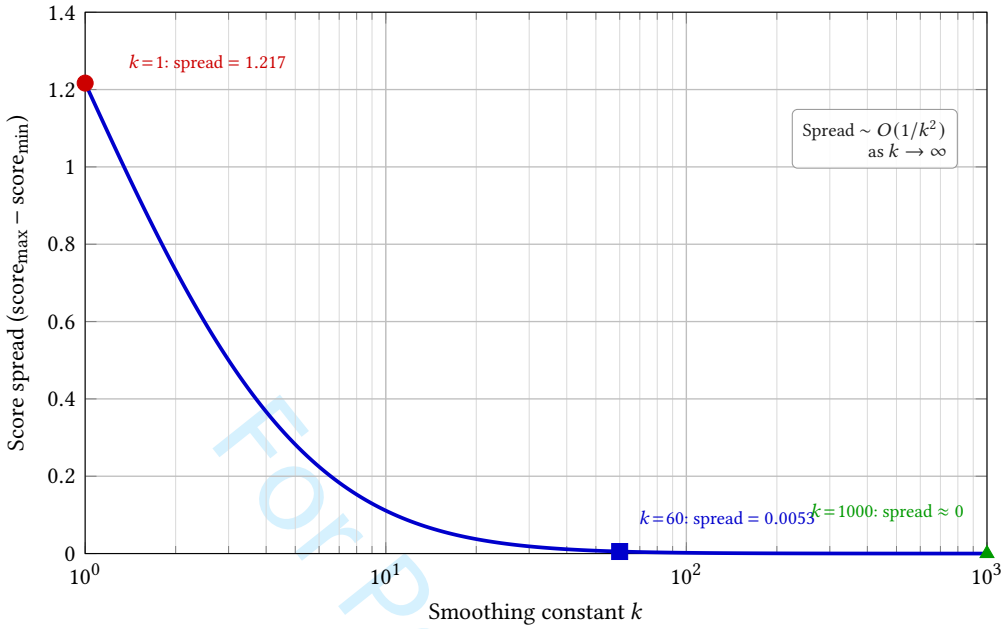

Fig. 6. Score spread (maximum score minus minimum score across the five example documents) as a function of the smoothing constant k, with equal weights $\boldsymbol{\alpha} = (1, 1, 1, 1)$. The spread collapses rapidly: from 1.217 at $k = 1$ to 0.0053 at $k = 60$ and 0.000022 at $k = 1000$. The three reference points from Table 8 are marked. Asymptotically, the spread decays as $O(1/k^2)$.

The standard choice $k = 60$ balances these concerns, preserving a meaningful score spread of 0.00530 while maintaining ranking agreement with the conservative regime.

9.4 Edge Cases

Single cross-reference ($|X(z_i)| = 1$). When a document has only one cross-reference, the min-rank operator is trivial: minrank = rank(x_1). The document loses the "safety net" of multiple cross-references. Under a uniform-rank model over N items, the expected minimum rank with m cross-references is $(N+1)/(m+1)$. For $N = 100$:

$$E[\text{minrank}] = \begin{cases} 50.5 & m = 1, \\ 33.7 & m = 2, \\ 25.3 & m = 3, \\ 16.8 & m = 5. \end{cases}$$

This systematic advantage for documents with more cross-references reflects a genuine information-theoretic property: more contextual connections provide more evidence for relevance.

No cross-references ($\alpha_X^{\text{sem}} = \alpha_X^{\text{lex}} = 0$). Setting both cross-reference weights to zero reduces the formula to standard two-ranker weighted RRF:

$$\text{score}(q, z_i) = \alpha_P^{\text{sem}} \cdot \frac{1}{k + r_P^{\text{sem}}(q, z_i)} + \alpha_P^{\text{lex}} \cdot \frac{1}{k + r_P^{\text{lex}}(q, z_i)}.$$

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Number of components in top-t (m)

Threshold t	$m = 1$	$m = 2$	$m = 3$	$m = 4$
1	0.0351	0.0453	0.0554	0.0656
2	0.0349	0.0448	0.0546	0.0645
3	0.0346	0.0442	0.0539	0.0635
5	0.0341	0.0433	0.0524	0.0615
10	0.0330	0.0411	0.0491	0.0571
15	0.0321	0.0392	0.0463	0.0533
20	0.0313	0.0375	0.0438	0.0500
30	0.0299	0.0347	0.0396	0.0444
50	0.0278	0.0307	0.0335	0.0364

Consensus beats single-expert Single-expert wins

Fig. 7. Consensus reward region for $k = 60$, $C = 4$ components with equal weights, and worst rank $R = 100$. Each cell shows the consensus score when m components place a document in the top t and the remaining $4 - m$ components assign rank R. Green cells indicate configurations where broad consensus across components outscores a single-expert document (rank 1 in one component, rank R in all others; single-expert score = 0.0351). Consensus with $m \geq 2$ components beats the single expert for all $t \leq 20$; with $m = 4$, consensus wins even at $t = 50$.

In our example with $\boldsymbol{\alpha} = (1, 1, 0, 0)$, the top three documents z_1, z_2, z_3 become nearly indistinguishable: their primary-only scores are $2/62 = 0.03226$, $1/63 + 1/61 = 0.03227$, and $1/61 + 1/63 = 0.03227$, respectively. The score spread among these three collapses to 0.00001. Reinstating cross-reference signals separates them: z_3 rises to a clear first (score 0.06505), while z_2 drops to fourth (0.06234). This demonstrates that cross-references provide discriminative power precisely where primary signals alone leave the ranking ambiguous.

Tied scores. Suppose all five documents share identical BM25 scores ($\mathrm{BM25}(q, z_i) = 7.0$ for all i). Under average-rank tie-breaking, each document receives $r_P^{\mathrm{lex}} = 3$, contributing $\frac{1}{k+3} = 1/63$ to every score. The lexical component becomes a constant offset and provides zero discrimination—the entire ranking is determined by the remaining three components. More generally, a tied component reduces the effective number of discriminating signals from four to three.

Taken together, these examples demonstrate that the four-component decomposition is neither a notational convenience nor an over-parameterisation. Each component plays a distinct role in ranking stability, and the interaction between weights, smoothing constant, and cross-reference structure produces behaviours that are precisely characterised by the formal properties established in Section 6. The component decomposition is visualised schematically in Figure 3, and the chain of theoretical extensions leading to the full formula is shown in Figure 8 and the commutative square of Figure 9. Figure 7 illustrates the consensus reward region from Corollary 6.23.

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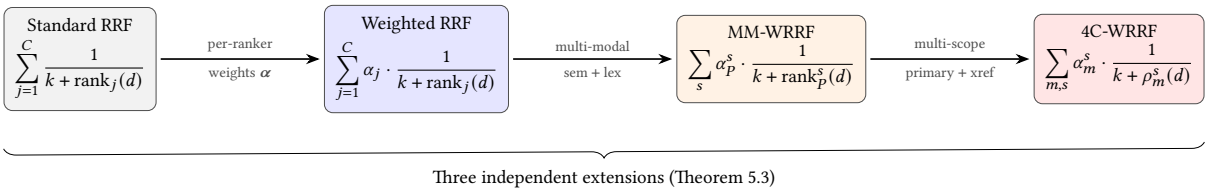

Fig. 8. The extension chain from standard Reciprocal Rank Fusion to the full four-component weighted formula. Each arrow denotes an independent extension: (i) per-ranker weighting introduces the weight vector $\boldsymbol{\alpha}$; (ii) multi-modal instantiation specialises the abstract rankers to semantic and lexical modalities; (iii) multi-scope expansion adds cross-reference signals with min-rank aggregation. By Theorem 5.3, these three extensions commute and their composition is unique up to reparameterisation.

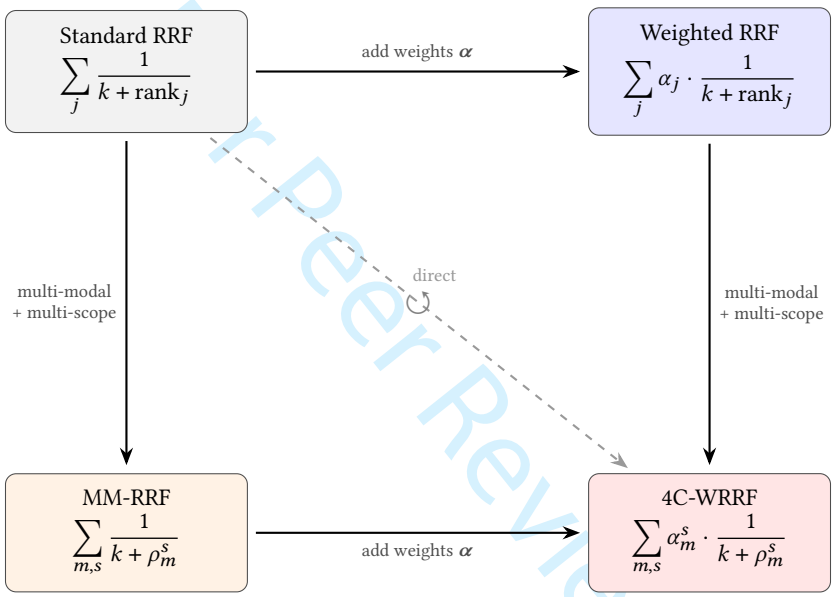

Fig. 9. Commutative square of extensions. The top path first introduces per-component weights (Standard RRF → Weighted RRF) then instantiates modalities and scopes (Weighted RRF → 4C-WRRF). The left path first instantiates the component structure (Standard RRF → MM-RRF, with unit weights) then introduces weights (MM-RRF → 4C-WRRF). Both paths yield the same formula; the dashed diagonal shows the direct one-step composition. Commutativity follows from the linearity of the score in the weight vector (Proposition 5.3).

10 DISCUSSION

10.1 The Role of the Harmonic Constraint

A natural concern is whether the harmonic constraint (H) introduced in Section 4 is an assumption or a consequence. We emphasise that it functions as a *selection principle*: among the infinitely many functions satisfying axioms A1–A5, the harmonic constraint uniquely picks out $f(r) = 1/(k + r)$. Remark 4.8 discusses why this selection is non-arbitrary: the harmonic kernel is the unique standard function satisfying the constant rank-score product property, and it is the form empirically validated by Cormack et al. [4].

One could replace the harmonic constraint with alternative selection principles—for instance, requiring exponential decay $f(r) = e^{-\beta r}$ or power-law decay $f(r) = r^{-\alpha}$—and the rest of the derivation (weighting, multi-modal extension, multi-scope extension with min-rank aggregation) would proceed analogously, yielding a four-component formula with a different kernel. The uniqueness theorem (Theorem 5.1) would then establish uniqueness relative to the alternative kernel. Our contribution is thus modular: the decomposition structure is independent of the specific kernel choice, while the kernel uniqueness result provides a principled default.

10.2 On the Absence of Experiments

This paper presents no retrieval experiments. This is a deliberate methodological choice, not an omission. Our contribution is formal and mathematical: we derive a formula, prove its uniqueness, establish its properties, and verify them mechanically. This approach follows the precedent of Fang, Tao, and Zhai [9], who established axiomatic foundations for retrieval functions without comparative experiments.

The empirical validation of RRF itself—including the choice $k = 60$—was thoroughly conducted by Cormack et al. [4]. Our work provides the theoretical scaffolding explaining *why* RRF works, rather than demonstrating *that* it works. The practical implications are that system designers can configure the four component weights and smoothing constant with confidence that the resulting formula satisfies all eight formal properties, and that no alternative formula satisfying the same axioms exists.

We note that the four-component decomposition is directly implementable in existing search infrastructure: Elasticsearch already supports weighted RRF with arbitrary ranker sets, and extending it to the four-component schema requires only enumerating the modality–scope pairs and specifying the weight vector $\boldsymbol{\alpha}$. An empirical study benchmarking the formula with learned weights on standard test collections (such as MS MARCO or BEIR) is a natural next step and is listed explicitly in our future work (Section 11).

10.3 Practical Implications

The formal properties established in Section 6 have direct practical consequences:

- **Scale invariance** (Theorem 6.15) means that RRF can fuse outputs from heterogeneous retrieval models (dense embeddings, BM25, cross-encoders) without score calibration or normalisation. This is a significant practical advantage over score-based fusion methods.
- **Lipschitz continuity** (Theorem 6.26) guarantees that weight tuning is stable: small perturbations in $\boldsymbol{\alpha}$ produce small changes in scores. For $k = 60$, the Lipschitz constant $L = 1/61 \approx 0.0164$ ensures that weight adjustments are smooth and predictable.
- **Consensus reward** (Theorem 6.22) formalises the intuition that documents performing well across all four retrieval signals are preferred over those excelling in only one. This property is particularly valuable in RAG settings, where a document with strong primary relevance *and* relevant cross-references is more likely to contain useful information.
- **Convexity / diminishing returns** (Theorem 6.11) explains why the smoothing constant k matters: it controls the emphasis placed on top-ranked documents versus lower-ranked ones. The diminishing returns property ensures that improving from rank 2 to rank 1 matters more than improving from rank 50 to rank 49, which aligns with user-focused evaluation metrics such as NDCG.

10.4 Extensions and Generalisations

The four-component decomposition naturally generalises along both axes:

- **Additional modalities.** Extending $\mathcal{M}$ to include cross-encoder reranking scores, BM25F variants, or other retrieval signals increases $|\mathcal{C}| = |\mathcal{M}| \times |\mathcal{S}|$ while preserving all axioms.

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- **Deeper scope hierarchies.** The binary scope set $\mathcal{S} = \{P, X\}$ could be extended to multi-hop cross-references ($\mathcal{S} = \{P, X^{(1)}, X^{(2)}, \ldots\}$), where $X^{(j)}$ represents j-hop citation-graph neighbours. The min-rank aggregation applies at each level.
- **Learned weights.** The weight vector $\boldsymbol{\alpha}$ can be treated as a learnable parameter optimised via learning-to-rank methods [12]. The Lipschitz continuity result ensures that gradient-based optimisation of $\boldsymbol{\alpha}$ is well-conditioned.
- **Alternative aggregation operators.** While Theorem 4.22 establishes min-rank as canonical under desiderata D1–D4, relaxing D2 (optimistic representativeness) admits max-rank or mean-rank alternatives for specific applications.

10.5 Limitations of the Verification

The Lean 4 verification covers five of eight properties: non-negativity, upper boundedness, monotonicity, convexity, and scale invariance. The remaining three—min-rank semilattice structure, consensus reward, and Lipschitz continuity—are not yet mechanically verified. The min-rank semilattice properties are straightforward but require lattice-theoretic infrastructure not available without Mathlib. The consensus reward theorem requires reasoning about specific numerical bounds. Lipschitz continuity requires real analysis. We regard the completion of these verifications, potentially using Mathlib, as future work.

11 CONCLUSION AND FUTURE WORK

We have presented the canonical decomposition of Reciprocal Rank Fusion into a four-component weighted scoring function for multi-modal, multi-scope retrieval. Starting from five axioms and a harmonic constraint, we derived the formula through three independent, commutative extensions and proved its uniqueness up to the smoothing constant k and the weight vector $\boldsymbol{\alpha}$. We established eight formal properties and verified five of them in the Lean 4 theorem prover. The connection to Condorcet voting theory provides additional interpretive depth, positioning the formula as a weighted multi-criteria positional scoring rule.

Future work. Several directions merit investigation:

(1) **Complete Lean verification.** Extending the machine-verified proofs to cover the remaining three properties (min-rank semilattice, consensus reward, Lipschitz continuity), potentially using Mathlib for lattice theory and real analysis.
(2) **Empirical validation of the decomposition.** While standard RRF has been extensively validated [4], the specific four-component decomposition with independently tuned weights has not been systematically benchmarked on standard test collections such as MS MARCO [13] or BEIR [17]. Ablation studies over the component weights on standard retrieval benchmarks would quantify the value of cross-reference signals.
(3) **Learned weight optimisation.** Treating $\boldsymbol{\alpha}$ as a learnable parameter and optimising it via gradient-based methods on labelled relevance data, leveraging the Lipschitz continuity guarantee for well-conditioned optimisation.
(4) **Alternative kernels.** Investigating whether power-law or exponential kernels yield different formal properties, and whether the four-component decomposition structure is preserved.
(5) **Dynamic k selection.** Developing principled methods for choosing or adapting k based on the corpus size and rank distribution, informed by the sensitivity analysis presented in Section 9.

The central message of this work is that the four-component weighted RRF formula is not an ad hoc construction but the unique canonical extension of standard RRF to the multi-modal, multi-scope retrieval setting. This theoretical foundation provides both explanatory power and design guidance for practitioners building hybrid retrieval systems.

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Received February 2026

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